Online Appendix to: Verified Detection and Prevention of Concurrency Anomalies in Multi-Agent Large Language Model Systems

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This supplement collects the raw experimental and benchmarking tables referenced from the main paper (Tables S1–S6) and the material moved out of it for length. Each table is referenced by its S-number from the main text; the surrounding analysis, findings, and interpretation remain in the main paper. Appendices G–S hold material moved out of the main paper for length; it is the main paper's own text, with its cross-references rewritten to point into the main paper or into this appendix. The full reproducibility artifact accompanies this appendix: `verus_count.sh` regenerates the obligation totals against the working tree, and `check_tables.sh` rescores every published cell of the main paper's Tables 3, 4 and 7 with an implementation of Definition 1 independent of the Verus-verified detector, failing non-zero on any disagreement.

APPENDIX A: CONCURRENT-SEMANTICS LIFT — FULL DEVELOPMENT

The four refinement proofs of the main paper hold under sequential interleaving: agent operations occur one at a time, atomically. This appendix develops the lift from sequential to concurrent semantics under an *abstract* mutex protocol.

What this lift now covers, and what it does not. Since the two guarded stores were rebuilt on `vstd`'s verified reader–writer lock, the deployed snapshot-isolation and pessimistic-locking stores assume nothing about a lock: their safety is established directly, on every state reachable under the verified lock, and `parking_lot::Mutex` is gone from both. The lift below therefore *does not describe a deployed guarded runtime*. It remains the mechanized statement of what an abstract lock must satisfy for the sequential refinement proofs to apply to a multi-threaded execution, and it is retained for that purpose. What still locks with `parking_lot::Mutex` is the agent client's pending-operation cell and the unguarded vanilla store, which is the L_0 baseline and carries no safety claim.

Atomic-event model. We define a concurrent operational semantics in which each runtime step is an atomic event of one of four kinds: `LockAcquire(a,c,t)`, `LockRelease(a,c,t)`, `Read(a,c,t,v)` and `Write(a,c,t,v)`, where a is the agent, c the cell, t a monotonic timestamp and v a value. A concurrent trace is a sequence of such events. The runtime state at any prefix is a ⟨holders, values⟩ pair: a map from cells to currently-holding agents (absent keys mean unlocked) and a map from cells to current values.

A step function `step(s,e)` transitions the state: `LockAcquire` adds $(c \mapsto a)$ to holders, `LockRelease` removes c, `Read` leaves s unchanged, `Write` updates `values[c] := v`. The companion enabling predicate `step_enabled(s,e)` encodes the abstract mutex protocol: `LockAcquire` requires the cell to be unlocked; `LockRelease` and accesses require the acting agent to currently hold the cell's lock; a `Read` must observe the cell's current value if one exists. A trace τ is well-formed iff every event in τ is enabled from its preceding state.

Verified theorems. The development (`lib_concurrent_semantics.rs`; 9 verified, 0 errors, 0 axioms) establishes: prefix monotonicity; mutex exclusion (at most one holder per cell); access implies lock held; reads observe the current value (no torn reads); distinct-agent access separation; and a helper that a holder change requires an intervening lock event.

Lift to sequential refinement. Distinct-agent access separation is the key result: per cell, the access events in a well-formed concurrent trace project to a strictly serialized sequence. The per-cell projection is exactly the sequential trace shape consumed by the refinement proofs, so any well-formed concurrent trace is observationally equivalent (per cell) to a sequential trace.

Scope, and the structural-tautology caveat. This lift has a structural-tautology character: well-formedness encodes the abstract protocol as a precondition on events, and the safety consequences follow from that encoding. It is not a verification of any concrete runtime; it is an identification, in mechanically-verified form, of the precise abstract protocol the sequential refinement proofs require.

Bounded weak-memory validation. While the abstract lift assumes sequential consistency, we additionally check the lock protocol under the RC11 weak-memory model [1] with the GenMC stateless model checker [2]. Encoding the acquire–read–generate–write–release discipline as N concurrent locked read–modify–writes on a shared cell, GenMC exhaustively reports no lost update—a lost update being exactly an A_1 witness—for

$N \in \{2,3,4\}$ (2, 6 and 24 complete executions). A relaxed-ordering control, with the acquire-CAS and release-store weakened to `memory_order_relaxed`, is reported as a data race with a concrete counterexample, confirming the release/acquire ordering is load-bearing. This does not extend to unbounded N, nor does it verify that any particular mutex implements the protocol.

APPENDIX B: RUSTBELT INTERFACE SPECIFICATION — FULL OBLIGATIONS

What this is and is not. The file `verus-detector/src/lib_rustbelt_interface.rs` is not a RustBelt proof. RustBelt [3] is a verified type system for Rust built on Iris; producing a RustBelt-grade verification of a mutex requires working inside Iris. What the file is: an explicit, mechanized specification of the obligations such a verification would discharge.

These obligations describe no deployed guarded runtime. Both guarded stores now run on `vstd`'s own verified reader–writer lock, so none of the obligations below is on the path of a deployed safety claim. They are retained as the statement of what the abstract lift of Appendix A requires of a lock.

The four obligations.

1) `lock_is_acquire`: acquiring the lock corresponds to the abstract `LockAcquire` event.
2) `drop_is_release`: dropping the guard corresponds to `LockRelease`; the borrow checker ensures the guard cannot outlive the critical section.
3) `event_seqcst`: the Rust memory model plus the LLVM pipeline preserve the total order of Acquire/Release events the event-level model assumes.
4) `no_panic_in_critical_section`: a property of the runtime source, not of any mutex. Unlike 1–3 this is not a bare stub: the same file proves 4a, panic-freedom of the modeled critical sections, and 4b, fail-safety under lock poisoning.

Obligation 4b is moreover no longer needed by the guarded stores: both install state only through the verified lock's release path, which requires the invariant, so a panic inside a critical section cannot leave a torn store. The residual stub count is therefore three, matching the trust-base accounting of the main paper. This appendix does not close the concurrency gap; it makes the residual auditable as three named obligations rather than a collective assumption.

APPENDIX C: MAST — TOPOLOGICAL SUSCEPTIBILITY AND DYNAMIC CONFIRMATION

Structural susceptibility across canonical topologies. The necessary condition for cross-agent A_1 has a purely structural part—two agents scheduled in one superstep where one reads a state channel the other writes—decidable from a graph's layers and per-node read/write channels alone, independent of any model output. This is `lib_l2_safety.rs::a1_witness` with the value inequality existentially relaxed, hence a sound over-approximation: a topology found non-susceptible cannot exhibit cross-agent A_1 at runtime for any input or model.

We applied it to sixteen canonical, publicly documented topologies—LangGraph supervisor fan-out, map-reduce with and without a reducer, orchestrator-workers, parallel tool calls, routing, reflection, evaluator-optimizer, plan-and-execute, hierarchical teams, collaboration, swarm, a memory chatbot, CrewAI sequential staging, multi-agent debate and the classic blackboard architecture—plus two controls. Six of sixteen are structurally susceptible: hierarchical teams, collaboration, swarm, blackboard, competing-agents consensus and a reducer-less map-reduce, in every case because concurrent agents read and write one shared mutable channel within a superstep. The other ten are immune by construction: disjoint result channels, an additive reducer, or strictly sequential staging. The contrast is sharpest between the reducer-based map-reduce (immune) and an identical workload with a read-modify-written shared accumulator (susceptible), isolating the reducer—not the workload—as the safety mechanism. The positive control is flagged and the negative control is not. The analyzer (`prevalence_static.py`) is deterministic and dependency-free.

The precision this bound requires, and what a framework supplies. The read and write channel sets above are declared, not recovered, and the distinction they support is fragile in a specific and measurable way. `python/langgraph_extract.py` builds the same four topologies as real LangGraph graphs, scores the same structural predicate over a 2×2 matrix of declared and extracted read and write sets, and reports:

read set / write set		susceptible
declared / declared	(the encoding above)	1 of 4
declared / extracted		3 of 4
extracted / declared		3 of 4
extracted / extracted	(a real front end)	3 of 4

Losing precision on either side collapses the split, so both are load-bearing—but the two sides are not alike. The read set is over-approximated by default, because a compiled node reports the whole state schema, and it is *recoverable*: annotating a node with `add_node(..., input_schema=S)` narrows what the framework reports to that node's declared channels. The write set is not recoverable at any annotation effort: LangGraph 1.2 has no per-node output schema and its `ChannelWrite` carries `static=None`, because the channels a node writes are whatever dictionary its body returns at run time. The obvious escape does not work either: the reducer flag (`BinaryOperatorAggregate` versus `LastValue`) is the one discriminating fact the compiled object does expose, but a last-value rule built on it reproduces the 1-of-4 split only when write sets are already known, and returns 3 of 4 otherwise—it needs the same missing input.

A shape the runtime refuses. Running the same graphs rather than only reading them adds one more constraint, and it cuts across the encoding. LangGraph rejects two co-superstep nodes writing one non-reducer channel at execution time: `LastValue.update` raises `InvalidUpdateError`, "At key '*c*': Can receive only one value per step." Of the four graphs above, three execute and the one the encoding calls susceptible does not. Every one of the six susceptible topologies in `prevalence_static.py` has two or more co-layer writers on the cell it is flagged for—`shared_scratch`, `messages`, `active_agent`, `blackboard`, `proposals`, `accumulator`—so for those of them intended as LangGraph graphs, the flagged shape is one the runtime will not run without a reducer, and a reducer is what the encoding calls immune.

This does not make A_1 vacuous in LangGraph, and the distinction matters. A reader sharing a channel with a *single* co-superstep writer is susceptible and does execute; that is the

deployable shape, and it is the one the main paper's live cross-agent stale-generation study measures. The structural bound and the live measurement are therefore about different shapes, and we say so rather than let the two zeros and the *k*-of-*N* figure be read as one body of evidence.

What an instrumented run recovers. The write set is the one input static extraction cannot supply, and it is precisely what a single instrumented execution observes, because a node's update dictionary *is* its write set. `prevalence_harness.instrument_layered` wraps every node of a graph, derives each node's superstep from the graph's own edges, and records an operation per node; on the three runnable graphs above the observed write sets matched the declared ones exactly. That is why the corpus instrument (`python/prevalence_corpus.py`) wraps nodes rather than reading the compiled object, and it is the method a third-party corpus study would use.

We therefore state the bound's scope rather than leave it to be discovered. The *k*-of-*N* figure characterizes *documented architectures under a declared encoding*. Turning it into a design-time analysis over third-party graphs requires one framework affordance that does not exist today and that is symmetric with one that does: a per-node write declaration to match the per-node `input_schema`. That is a concrete request to framework authors, and it is the gating item for the static robustness procedure the main paper's future-work section describes, rather than an engineering detail of it.

Dynamic confirmation, and exactly what it confirms. Each susceptible topology is instantiated in its unconditionally-concurrent form in a small BSP/superstep executor in which each node is a real model call (`prevalence_dynamic_run.py`), and all nine instantiations were run for $n = 20$ trials on each of the two model families whose 360 session records the artifact carries: `gpt-4o-mini` and `claude-haiku-4-5`. The outcome is identical on both: every structurally-immune topology and the negative control fired in $0/20$, while every susceptible instantiation and the positive control fired in $20/20$ (exact 95 % Clopper–Pearson $[0.83, 1.00]$). Scored with the *superstep* form of the predicate—a co-superstep write of a different value by another agent—since each round is a single superstep whose commits land at one tick.

Two limits of that run are load-bearing and we state them rather than let the $20/20$ carry more than it can. First, the superstep form is a *different* relaxation of Definition 1 from the structural predicate above: the structural analyzer drops the value conjunct and keeps the ordering, while the executor keeps the value conjunct and drops the window. Because the executor stamps each record's read and commit as consecutive ticks, all 1,320 committed records have window width one, and rescoring the same records under the full Definition 1 yields $0/360$—the identical width-one limitation the cookbook-derived and MAST-Data tables below declare for their own corpora. Second, the executor pre-populates every cell with a sentinel so that first-layer reads are recorded at all, and all 520 superstep firings are reads of that sentinel; none is an agent reading a value another agent produced. Both facts are recomputable from the committed records in `python/dynamic_oprecords`.

What the run therefore establishes is that a superstep executor does schedule the structurally susceptible topologies concurrently on a shared channel, and the immune ones not—the structural precondition exercised rather than inferred. It is not an independent measurement of value-conflicting A_1, and the main paper does not use it as one. This demarcates the *structural* scope of A_1 rather than estimating prevalence: absent by construction from reducer- and fan-out-structured deployments, structurally present in shared-mutable-cell designs run concurrently. The value-conflicting measurement is the live study of the main paper's Section 6.1, where the window is real.

Superseded as a prevalence instrument. The main paper's Section 6.8 replaces this bound with a static census of public repositories, which recovers per-node write sets from source rather than asking the framework for them: 407 graph sites in 98 sampled repositories, none with a confirmed co-scheduled reader and writer of a last-value channel.

APPENDIX D: LOCATING CONTEMPORARY RUNTIMES ON THE CHAIN — FULL PLACEMENTS

Offered as illustrative discussion, not a formal contribution. Each placement is approximate, based on published descriptions rather than source analysis, and conditional on workload assumptions made explicit per runtime. We report each runtime's avoidance set as a subset of $\{A_1, A_2, A_3, A_6\}$, naming a chain level only when the set is on-chain.

SagaLLM [4]: avoidance set $\{A_3\}$ (off-chain) on compensable workloads. Saga-style compensation addresses A_3 by retracting external effects when their basis is invalidated (conditional on reversibility). It does not address A_1 (no read-set tracking), A_2 (no registry stability) or A_6 (no intra-operation write atomicity). The set is $\{A_3\}$ on compensable workloads, $\emptyset$ (L_0) where effects are irreversible, and $\{A_1, A_3\}$ (L_2) with a complementary OCC layer.

Atomix [5]: avoidance set varies by workload. Atomix tags tool calls with epochs, tracks per-resource frontiers and commits only when progress predicates indicate safety; the frontier-gated commit prevents intra-operation reordering ($\neg A_6$) and compensation handles retraction ($\neg A_3$, modulo irreversibility). A_1 prevention depends on whether read-sets are frontier-tracked: yes for explicit dependencies, no for implicit. The set is $\{A_1, A_3, A_6\}$ (L_3, on-chain) on explicit-dependency compensable workloads, and $\{A_3, A_6\}$ or $\{A_6\}$ (off-chain) otherwise.

CodeCRDT [6]: avoidance set $\emptyset$ (L_0) under our model. CRDT merging incorporates every concurrent write into the final state—a different contract from ours. Under our single-store, global-commit-log model, merging admits A_1. This is not a quality ranking: merge-convergence is an axis orthogonal to the serializability lattice, and reading L_0 as "worst" is a category error. Supported by two TLC runs: `MC_CodeCRDT_AdmitsA1` exhibits the witness (2,997 states, depth 5), and `MC_CodeCRDT_RYW` explores read-your-writes without a violation (exhaustive at small bounds: 9,348,770 states; partial at $|A| = 3$ up to 1.52×10^8)—the latter bounded, not a proof.

What the placements show. Of five distinct placements, two are on the chosen chain and three are off-chain. That is itself informative: the chain cannot name every guarantee real runtimes provide, which is why the lattice is primary and the chain one linearization. None reaches L_4.

APPENDIX E: REFERENCE RUST RUNTIME — FULL SAFETY PROOFS

The reference crate `mac-consistency-runtime` implements the three runtimes (vanilla, pessimistic locking, snapshot isolation

with an optional SSI mode) over a common `Store` trait, emitting `OpRecord` traces classified by the ported verified detectors. Its integration tests reproduce the behaviors of the main paper: vanilla edit-review produces an A_1 witness; pessimistic locking blocks at `begin` and produces a clean trace; snapshot isolation aborts on validation when committing writes; default-SI misses the no-write stale-read pattern—the silent residual of 5% on edit-review and 8% on triage reported in the main paper's Table 4—and SSI mode closes it.

Which code is verified. Four components of the crate are the verified bytes themselves, compiled by plain `cargo` with proofs erased: `si_concurrent.rs` and `pess_concurrent.rs`, byte-identical to their `verus-detector` originals and carrying the deployed guarded stores; and `l2_exec.rs`, byte-identical to `lib_l2_exec.rs`, which carries the L_2 runtime and `can_commit`, a decision procedure verified two-sided against `commit_valid`; and `l4_exec.rs`, byte-identical to `lib_l4_exec.rs`, which carries the validated dispatcher of L_4. The vanilla store and the agent client are not verified and carry no safety claim.

Verification status. The abstract state machines of the three strategies verify under Verus with forty obligations closed across three files (`lib.rs`, 23, pessimistic; `lib_ssi.rs`, 8, SSI; `lib_default_si.rs`, 9, default-SI), zero errors, zero `assume` in proof bodies, zero `external_body` in safety-theorem bodies and zero added axioms.

Pessimistic safety theorem. Under sequential interleavings of atomic `begin_step` and `commit_step` transitions,

`theorem_pessimistic_prevents_a1`:
$\forall s.\ \mathit{reachable}(s) \Rightarrow \neg A_1(s.\mathit{trace})$.

The proof maintains seven inductive invariants. Four decompose the natural two-conjunct lock-uniqueness predicate into single-conjunct foralls; the decomposition is forced by SMT trigger matching, which is unreliable on the conjoined form and discharges cleanly on the splits. Reachability is expressed over state sequences rather than as a recursive predicate, since the recursive form fails to unfold under Verus's fact-mode resolution.

SSI safety theorem. Under sequential interleavings of `ssi_begin_step`, `ssi_commit_success_step` and `ssi_commit_abort_step` with `validate_no_write = true`,

`theorem_ssi_prevents_a1`:
$\forall s.\ \mathit{reachable}(s) \Rightarrow \neg A_1(s.\mathit{trace})$.

Seven invariants, of which the load-bearing one is `inv_no_intervening_write`: for records r_i, r_j and cell c, c does not simultaneously lie in $r_i.\mathit{read_set}$ and $r_j.\mathit{write_set}$ with $r_i.\mathit{read_time} < r_j.\mathit{write_time} < r_i.\mathit{write_time}$.

Default-SI conditional safety theorem. The bypass transition is the no-write gap: it appends a record with empty write-set without revalidating reads. The conditional theorem is

$$\forall s.\ \big(\mathit{reachable}(s) \wedge \mathit{all_records_have_writes}(s.\mathit{trace})\big) \Rightarrow \neg A_1(s.\mathit{trace}).$$

It is not an unconditional safety theorem: default-SI admits A_1 by design when a caller exercises the bypass under contention, which is the measured residual of 5% on edit-review and 8% on triage. It is a Verus characterization of the workload class on which default-SI is safe—those in which the agent layer never issues a no-write commit. That class is exactly the one in which the main paper measures no firing at all: plan-execute, where the pooled rate across three runs is $0/300$. The design rationale is the `validate_no_write` toggle: a deployer who cannot guarantee the all-writers workload runs SSI; one who can may run default-SI as an optimization.

Scope, and what the refinements cover. The three theorems establish $\neg A_1$ for traces produced by the three abstract state machines: unconditionally for pessimistic and SSI, conditionally for default-SI. Vanilla is not covered; it admits A_1 by design and is the control.

The four spec–runtime refinement proofs relate those abstract machines to *models of the representations the runtime shipped before the guarded stores were rebuilt on the verified lock*—the `BTreeMap` version chain and the `HashMap` hold table—not to the flat version chain and hold table the stores deploy today. The safety of the deployed guarded stores does not rest on them: it is established directly, as the absence of a stale-generation window on every state reachable under the verified lock. The main paper's Theorem L1j closes the remaining step on the emission side, proving that a faithful emission of a window-free store trace contains no Definition-1 witness; refining the two concurrent stores to the abstract machines, so that both routes meet, remains open.

APPENDIX F: TLAPS COHERENCE CHECK AND THE A_1 GENERATION-PHASE LOWER BOUND

The chain definitions are subjected to a mechanical consistency check in TLAPS (`Hierarchy.tla`). The check is shallow by design: it verifies that the textual conjunction structure of the level predicates yields the implications $L_{i+1}(h) \Rightarrow L_i(h)$ we claim, and that the definitions contain no typo or accidental redundancy. It is not a substantive verification result, and we state the limitation up front because the substantive mechanization is the Verus chain of the main paper.

Eleven theorems are stated: four adjacent-pair containments, six transitive soundness results, and one aggregate. TLAPS decomposes them into 15 atomic obligations, all discharged by definitional unfolding.

A second result: same-agent exclusion, stated as an equivalence. Writing $\mathit{ReadSetLock}(h,i)$ for "no operation commits a write to a cell of $h[i].\mathit{read_set}$ within the window $(h[i].\mathit{read_time}, h[i].\mathit{write_time})$" and $\mathit{ValueAgreement}(h,i)$ for "every such intervening write matches $h[i]$'s recorded read value", `A1LowerBound.tla` proves in TLAPS (28 atomic obligations, no `assume` and no `omitted`):

> For every $h \in \mathit{Seq}(\mathit{OpRecord})$ satisfying the two well-formedness invariants below, $\neg\textsc{StaleGeneration}(h)$ implies $\forall i :\ \mathit{ReadSetLock}(h,i) \vee \mathit{ValueAgreement}(h,i)$.

An earlier version of this appendix called that a generation-phase lower bound that constrains every stale-generation-free execution. It is not, and `tla/MC_A1LowerBound.tla` with `tla/check_a1lowerbound.sh` settles what it is, by TLC over 83,233 traces at $|\mathit{Agents}| = 2$, $|\mathit{Cells}| = 1$, $|\mathit{Values}| = 1$, length ≤ 2 and times in 0..2. Three facts.

First, the disjunction is degenerate: $\mathit{ReadSetLock}(h,i) \Rightarrow \mathit{ValueAgreement}(h,i)$ holds on every trace with no hypothesis at all, because when no intervening write meets the read set, *ValueAgreement*'s inner universal ranges over the empty set. So the conclusion is *ValueAgreement* alone and there is one

mechanism, not a choice of two. Second, the implication runs both ways—$\forall i\, ValueAgreement(h,i) \Rightarrow \neg$STALEGENERATION$(h)$, and that direction needs *neither* hypothesis—so the result is an equivalence with asymmetric hypotheses rather than a bound. A statement equivalent to the definition does not constrain executions the definition does not; it says under which condition one exists.

Third, the hypotheses are load-bearing, and for exactly one reason. (i) $NonOverlappingAgent(h)$: no two operations of the same agent overlap in $[read_time, write_time]$. (ii) $MonotonicOp(h)$: every operation reads no later than it writes. Dropping either yields a TLC counterexample, and in both the witness is a single agent conflicting with *itself*—a configuration STALEGENERATION cannot witness, since it requires $h[i].agent \neq h[j].agent$. Excluding that case is what the proof's step $\langle 1 \rangle 5$ does and is the theorem's whole content. We therefore describe the result as a same-agent exclusion lemma, which is what it is, rather than as a bound on executions.

A third result: L_1 is commit-order serializability. `CommitOrder.tla` proves the main paper's Theorem 1 in TLAPS (60 obligations, no omitted step; `tlapm -toolbox 0 0 CommitOrder.tla`, log `formal/tla/tlaps.CommitOrder.log`, 2026-10-14: "All 60 obligations proved"): under the store axiom, read-before-commit and per-agent non-overlap, a history with no STALEGENERATION witness is serializable in commit order. `check_commitorder.sh` runs fifteen TLC claims that pin down what the statement says: it holds over all 35,659 histories of length at most three, is not vacuous, fails when either droppable hypothesis is removed, its relational definitions equal `Memory.tla`'s `CHOOSE`-based helper and a literal serial replay, its converse fails (a value overwritten and then restored), STALEGENERATION also fires on a history serializable in another order, and its three hypotheses are invariants of `Memory.tla` over 2,961,434 reachable states. The shared definitions are byte-identical across the proof module and the two TLC harnesses, and the round's script diffs them, so TLC cannot silently check other definitions than the ones TLAPS proved about.

RAW EXPERIMENTAL AND BENCHMARKING TABLES

TABLE S1

Per-session token cost by runtime and workload, captured from OpenAI usage callbacks across 900 sessions (100 per cell). Pricing: $2.50/M input, $10.00/M output. 95 % bootstrap CIs (1,000 resamples). Cost in milli-dollars. **These are per-session costs.** The three runtimes do not complete the same work in a session, so the main paper re-prices them per *sound* committed operation in its Finding 6; read that table alongside this one.

Runtime	edit-review	plan-execute	triage
Vanilla	$3.13 m [3.11, 3.16]	$1.93 m [1.88, 1.98]	$3.23 m [2.93, 3.55]
Pessimistic	$3.28 m [3.21, 3.38]	$1.91 m [1.85, 1.97]	$5.21 m [4.90, 5.54]
Snapshot iso.	$3.11 m [3.09, 3.13]	$1.95 m [1.91, 1.99]	$3.13 m [2.83, 3.46]

TABLE S2

Per-session token cost, Anthropic usage fields, 900 sessions. Pricing: $3.00/M input, $15.00/M output. Same per-session caveat as Table S1.

Runtime	edit-review	plan-execute	triage
Vanilla	$16.21 m [16.14, 16.29]	$9.44 m [9.41, 9.47]	$12.36 m [12.33, 12.38]
Pessimistic	$16.43 m [16.39, 16.47]	$9.44 m [9.41, 9.47]	$27.73 m [25.39, 30.33]
Snapshot iso.	$16.25 m [16.18, 16.33]	$9.44 m [9.41, 9.47]	$12.34 m [12.31, 12.36]

TABLE S3

Cookbook-derived sessions: the structural *precondition* of A_1, not A_1 itself. 600 sessions, gpt-4o, AutoGen `RoundRobinGroupChat`, unmodified framework. The verified detector fires in $0/600$: the adapter stamps each record's read and commit as consecutive ticks, so every window has width one and Definition 1 cannot fire on any of them. What these columns report is the precondition—a slot one agent read is later rewritten with a different value by a different agent. Five stateless scenarios show none across 500 sessions; the typed-shared-state scenario shows it in $89/100$.

Scenario	precondition	cross-agent	mean events
`research_collab`	0.0 % [0, 3.6]	0.0 % [0, 3.6]	5.7
`supervisor`	0.0 % [0, 3.6]	0.0 % [0, 3.6]	6.8
`hierarchical`	0.0 % [0, 3.6]	0.0 % [0, 3.6]	7.5
`code_review`	0.0 % [0, 3.6]	0.0 % [0, 3.6]	5.6
`customer_triage`	0.0 % [0, 3.6]	0.0 % [0, 3.6]	7.0
`shared_workspace`	89.0 % [81.2, 94.4]	89.0 % [81.2, 94.4]	9.1
Definition-1 firings, all scenarios: $0/600$*, by construction.*			

TABLE S4

MAST-Data projection by framework. The zero is *by construction*, not a measurement: the adapter stamps each record's read and commit as consecutive steps, so every window has width one and Definition 1 cannot fire. The corpus therefore cannot show A_1 either way; what the projection measures is a structural audit of which architectures expose the read-before-write pattern at all. AppWorld's task-block format was not parsed by the generic fallback and is excluded from the denominator rather than imputed.

Framework	n_{parsed}	n_{total}	events/sess	Def.-1 firings
ChatDev	130	130	18.2	0 (width-one)
MetaGPT	230	230	1.0	0 (width-one)
Magentic	195	195	11.8	0 (width-one)
OpenManus	30	30	2.9	0 (width-one)
HyperAgent	2	30	34.5	0 (width-one)
AG2	13	597	1.4	0 (width-one)
AppWorld	0	30	—	—
Total	600	1,242	—	0 (width-one)

TABLE S5

The aborts in this study are injected at fixed probabilities (pessimistic 0.20, SSI 0.05) and the stores do not run; it prices a retry, not a discipline (Appendix O; wall-clock on the real stores is the main paper's Section 6.6). Within-model (gpt-4o) wall-clock cost by workload and strategy, 270 sessions. All pessimistic/SSI intervals overlap the vanilla interval: no strategy overhead is statistically separable from API-latency noise at $n = 30$. "Ab." is total aborts across the 30-session cell.

Workload	Strategy	wall-clock (s)	95 % CI	Ab.
edit-review	vanilla	9.16	[8.31, 10.03]	0
	pessimistic	8.24	[7.22, 9.41]	7
	SSI	7.01	[6.43, 7.71]	0
plan-execute	vanilla	6.49	[5.61, 7.75]	0
	pessimistic	7.44	[5.60, 10.19]	6
	SSI	6.76	[5.88, 7.97]	0
triage	vanilla	4.81	[4.57, 5.07]	0
	pessimistic	5.12	[4.82, 5.49]	14
	SSI	5.74	[4.97, 7.10]	1

TABLE S6
The aborts in this study are injected at fixed probabilities (pessimistic 0.20, SSI 0.05) and the stores do not run; it prices a retry, not a discipline (Appendix O; wall-clock on the real stores is the main paper's Section 6.6). Open-weights (Llama-3.2 via Ollama) within-model wall-clock cost, 270 sessions. All pessimistic/SSI intervals overlap vanilla; point ratios lie in $[0.82, 1.04]\times$.

Workload	Strategy	wall-clock (s)	95 % CI	Ab.
edit-review	vanilla	38.44	[34.51, 43.57]	0
	pessimistic	35.67	[33.45, 38.24]	23
	SSI	34.35	[33.23, 35.55]	5
plan-execute	vanilla	30.28	[25.38, 35.32]	0
	pessimistic	30.68	[25.31, 36.14]	11
	SSI	24.75	[20.50, 29.37]	2
triage	vanilla	25.54	[21.64, 29.41]	0
	pessimistic	26.49	[23.01, 29.55]	23
	SSI	23.29	[19.84, 26.79]	5

APPENDIX G: VERUS DEVELOPMENT: HELPER LEMMAS, GATE LEMMAS AND THE COMBINED SCORECARD

Flat-trace residue: what it detects, and what it does not.: A flat operation history that exposes neither *preds* nor *aborted*—a black-box trace from an un-instrumented runtime—cannot decide Definition 4 of the main paper directly. For that setting we retain the *residue* predicate

$$\begin{aligned} \mathit{Residue}(h) \;\equiv\; \exists\, j,\, \exists c \in h[j].\mathit{read_set} :\\ v = h[j].\mathit{read_values}[c] \neq \mathrm{NULL} \;\wedge\; \forall k \neq j :\\ \neg\big(c \in h[k].\mathit{write_set} \;\wedge\; h[k].\mathit{write_time} \leq h[j].\mathit{read_time} \;\wedge\\ h[k].\mathit{write_values}[c] = v\big), \end{aligned}$$

evaluated on the *projected* flat history: exactly the reads and writes of committed, unaborted operations, with their values and times, and nothing else. Earlier versions of this paper used that phrase without defining it and asserted $A_3 \Rightarrow \mathit{Residue}$ on it. The implication is false, and five mechanized results (`lib_a3_residue.rs`, 6 verified, 0 errors) replace it. `projects` states the projection; then:

- **R1, not sound in general.** A surviving writer of the *same* value before the read hides the aborted writer's value. A three-operation history satisfies the projection and the cascade while the residue stays silent (`lemma_residue_misses_a_cascade_on_repeated_values`).
- **R2, sound under unique writes.** If no two committed operations write the same value to a cell, a surviving reader of an aborted writer always leaves a residue witness (`lemma_residue_sound_for_direct_cascades_under_unique_writes`). This is the assumption black-box checkers already make: Elle requires uniquely identifiable writes [10] and Cobra's inference keys on write identity [72].
- **R3, chains.** The same holds along a chain of reads ending at an aborted writer—the causal-closure form—through its first aborted link (`lemma_residue_sound_for_cascade_chains_under_unique_writes`).
- **R4, not complete.** A read of a value no logged operation wrote leaves a residue with no cascade at all (`lemma_residue_fires_without_a_cascade`); a deployment that logs cell initialization as a write removes this class.
- **R5, blind to relabeled effects.** When a cascade flags a dependent *after* its effects are out, the projection drops that operation and the residue sees nothing, while Definition 4 of the main paper fires (`lemma_residue_blind_to_relabeled_effects`). Section 6.3 of the main paper counts this class on a runnable arm: 767 to 1,288 dependents per 1,000 scenarios under the superseded design.

The residue is therefore a screen for the superseded surviving-dependent form under a uniqueness assumption, not a sound over-approximation of Definition 4 of the main paper. A deployment that needs the cataloged predicate must expose provenance and externalization, as the runtimes of Section 5.3 of the main paper do.

Helpers (sound and complete). The primitive predicates and lookup operations are verified as exec–spec equivalences:

- `reads(op, c)` returns `true` iff $\exists k.\ 0 \leq k < |op.\mathit{read_set}| \wedge op.\mathit{read_set}[k] = c$;
- `writes(op, c)` satisfies the analogous equivalence over the write set;
- `contains_tool(v, t)` satisfies the analogous equivalence over a tool sequence;
- `first_value_exec(s, c)` agrees with a deterministic first-match spec function *first_value* over $\mathrm{Seq}(\mathit{CellId} \times \mathit{Value})$, supported by a uniqueness lemma showing that the first-match index is unique.

The first-match uniqueness lemma is required to make the spec-level value lookup well-defined under Verus's `choose` semantics: without it, the existential witness used to define *first_value* could be any matching index, breaking exec–spec agreement.

Refinement structure. For each strategy we define a concrete-side state record mirroring the relevant fields of the deployed Rust `Inner` struct (`data`, `locks`, `agent_holds`, `clock`, `trace` for pessimistic; `store`, `last_write`, `clock`, `trace` for SSI), an abstraction function α mapping concrete states to instances of the abstract state machine of Section 5.2 of the main paper, and concrete-side transitions corresponding to each abstract transition. Each refinement lemma has the shape

$$\mathrm{concrete_step}(c, c') \implies \mathrm{abstract_step}(\alpha(c), \alpha(c')),$$

together with an initial-state correspondence $\alpha(c_0) = a_0$. Composed with the safety theorems of Section 5.2 of the main paper, this establishes that every concrete execution maps to an abstract execution satisfying $\neg A_1$.

The four refinements. Each refinement discharges, for its strategy's concrete transitions (begin/commit and, for SSI/default-SI, abort), an initial-state correspondence and the structural helpers—Map-insert, snapshot-build, trace-push, and lock-release commutativity—that lift each concrete step to its abstract image; the per-helper detail is in the reproducible artifact. The counts are 31 (pessimistic), 18 (SSI projection), 19 (SSI version-chain), and 18 (default-SI). Two points are load-bearing rather than mechanical. The version-chain refinement (`lib_refinement_ssi_chain.rs`) targets the literal deployed `BTreeMap<CellId, Vec<(Time,Value)>>` and takes chain-monotonicity (every chain non-empty with strictly increasing times bounded by *clock*) as a precondition; we discharge its preservation across all reachable states (`lib_si_commit_invariant.rs`: 4 verified, no `assume`/`admit`), so the invariant holds by induction rather than by assumption. The default-SI refinement (`lib_refinement_default_si.rs`) covers the bypass branch (empty write-set, validation skipped) unconditionally—the refinement itself imposes no workload restriction; only the safety theorem does—and the new technical piece is that an empty concrete write-set maps to an empty abstract one under the *Set* :: *map* abstraction.

In-the-loop verification of the SI validation gate. For the snapshot-isolation validation decision we go beyond refinement to direct execution-level verification: we lift the deployed `SnapshotIsolationStore::commit` read-set check into a Verus `exec` function, prove it sound and complete against the freshness predicate (`validate`: 5 verified, no `assume`/`admit`/`external_body`), and the deployed commit now invokes that exact function (confirmed by the unchanged test suite). The procedure that selects commit versus abort at run time is therefore the proven function itself; the unverified remainder is only the marshalling that flattens the version map into the gate's input; since the integration of Section 5.2 of the main paper, the gate runs inside the verified concurrent store, under `vstd`'s verified lock

rather than a `parking_lot::Mutex`. The A_1 detector is verified in the same `exec` style (9 verified). For these two procedures the spec–runtime residual is reduced from a refinement argument to a marshalling correspondence: the executable code that runs is the verified code.

Pessimistic acquisition: gate and exclusivity invariant. The pessimistic enforcer is verified to the same depth: its begin-time acquisition check is an `exec` function proved sound and complete against the no-foreign-holder predicate (`can_acquire`: 5 verified), and we prove that acquisition and release preserve consistency of the two holder views (`lib_pessimistic_invariant.rs`: 3 verified, no `assume`/`admit`), so each cell has at most one holder in every reachable state—the property that excludes concurrent same-cell operation and hence A_1 at this layer. Both deployed enforcers thus carry an exec-verified decision gate and a mechanically-proved state-machine invariant, not only a refinement to an abstract model.

Trust base: two foundational axioms. The four refinement files share a trust base of two axioms, disclosed here in full:

- `axiom_string_to_int_injective`: $\forall s_1, s_2.\ \mathit{string_to_int}(s_1) = \mathit{string_to_int}(s_2) \Rightarrow s_1 = s_2$. This formalizes that the abstraction function for string identifiers (cell IDs, agent IDs, value strings) is injective. In the deployed runtime, identifiers are arbitrary `String`s and the corresponding abstract type is $\mathbb{Z}$; injectivity of any well-defined encoding is required for the abstraction to commute with Map-insertion. This is a structural assumption about the alphabet, not a domain claim.
- `axiom_null_sentinel`: $\mathit{value_alpha}(\langle\text{'N','U','L','L'}\rangle) = 0$. The concrete runtime uses the string `"NULL"` as a sentinel for absent values returned from reads on uninitialized cells; the abstract specification uses the integer 0 as the corresponding null. The axiom asserts that these correspond under the abstraction function. This is a one-line constant-folding assumption.

Both are structural properties of the encoding alphabet, asserting nothing about the lattice, catalog, or strategies (see the consolidated trust base of Section 5.2 of the main paper). A natural candidate for a third axiom—$\mathit{finite}(s) \Rightarrow \mathit{finite}(s.\mathrm{map}(f))$, used for the finite-domain conjunct of the abstract commit step—is instead a proven Verus lemma (`lemma_set_map_preserves_finite`, by induction on $|s|$), keeping it out of the trust base.

Relating the projection and chain refinements. The projection refinement (latest version per cell) and the chain refinement (full $\mathit{Seq}(\mathit{Time}, \mathit{Value})$ history) refine to the same abstract SSI machine: the projection is the minimal abstraction sufficient for the safety theorem, the chain is the literal deployed Rust shape. Distributing both makes that trade-off explicit.

Combined formal scorecard. The Verus mechanization comprises (all counts live-confirmed by `verus_count.sh`; all zero `assume`, zero `admit`): 24 for the detector exec–spec equivalence chain (Section 5.1 of the main paper); 40 across the three strategy safety theorems; 86 across the four refinement files (under the two shared axioms); 9 for the concurrent-semantics lift (Section 5.2 of the main paper); 33 for the L_2 safety theorem (Section 5.3 of the main paper, with *pred_closed* proved inductive and Theorems L_2g/i/i$^+$); 40 net for the exec-mode L_2 runtime (Section 6.3 of the main paper, on natively-injective `u64` interning, no added axiom, and now carrying `can_commit`, a decision procedure verified two-sided against `commit_valid`; its self-contained file verifies at 73 end to end, and every entry point checking its own precondition); 9 for the A_4 formalization (Appendix J); 4 for the RustBelt interface (Section 5.2 of the main paper, including panic-freedom and poisoning fail-safety, with the 3 `RUSTBELT_OBLIGATION` stubs); 6 for the L_3 saga theorem and 5 for the commit-order sequencer (Theorem L_3g); 9 for the L_4 validation-at-dispatch theorem; 6 for the A_3 residue characterization (`lib_a3_residue.rs`, Appendix G); 3 for the L_2 state-to-trace projection (Theorem L_2j); 7 for the LangGraph reducer-channel theorem; and 8 for the ETag/OCC read-validation channel (Appendix Q); 5 for the joint-composition root (`lib_consistency_lattice.rs`, beyond its three re-included component modules, Section 5.4 of the main paper); 7 for the exec-mode L_3 sequencer (`lib_l3_exec.rs`); 11 for the exec-mode L_4 dispatcher (`lib_l4_exec.rs`, Section 6.3 of the main paper); 18 for the concurrent SI store over the verified reader–writer lock (`lib_si_concurrent.rs`); and 31 for the concurrent pessimistic store over the same lock (`lib_pess_concurrent.rs`), both in Section 5.2 of the main paper; and 3 for the L_1 projection (`lib_si_projection.rs`, Theorem L1j in the same section), which transports the concurrent SI store's structural guarantee to a Definition 1 of the main paper guarantee on the emitted history. Finally 7 for the basis ledger model (`lib_l5_ledger.rs`, Appendix U), counted as scaffolding. **Total: 371 distinct verified obligations** (0 errors), all unconditional under the two foundational axioms; the stochastic case is left open at the specification level (Appendix I) and the artifact carries no probabilistic obligation or axiom. Every named lattice point L_0–L_4 is backed by a mechanically-verified realizing runtime model: deployed and refined to Rust at L_0/L_1; at L_2 and L_4 the verified exec-mode bytes themselves are executed and measured (A_3: $0/1000$ versus 519–$624/1000$ for a baseline that releases at commit; A_2: $0/1000$ versus 387–$630/1000$ for live and for snapshot resolution alike); L_3 is exec-verified and measured on a separate twin (A_6: $0/1000$ versus 487–$1000/1000$). Counting every verified source file and subtracting module/textual re-inclusion, the development verifies **392 distinct obligations**; the 21 between the 371 curated headline and 392 are independent invariant lemmas and exec runtimes (`lib_detect_a1_exec.rs` 9, `lib_pessimistic_exec.rs` 5, `lib_pessimistic_invariant.rs` 3, `lib_si_commit_invariant.rs` 4) that support the named results without being lattice points. `verus_count.sh` regenerates both figures (371 curated; 392 with `-full`), measuring the working tree and printing its revision.

APPENDIX H: CONCURRENT STORES: WHAT THE OPEN REFINEMENT STEP MUST SAY, AND WHAT THE VERIFIED LOCK COSTS

Three properties distinguish this from Theorem L2j and we state them rather than let the shared name imply more. First, the emission relation is *index-preserving and total*—one `OpRecord` per store record, in order, filtering nothing—which is what `SiStore` does, since it appends exactly one record per commit. Theorem L2j's relation instead quantifies over committed-clean transactions and therefore presupposes that the emitter has already dropped every aborted record. Second, the cell conjuncts are containments rather than equalities, so an emitter that publishes fewer cells than its record holds is still faithful; that weakens the hypothesis and

strengthens the theorem, at the price of admitting a degenerate emitter that publishes nothing and is trivially clean. The theorem therefore bounds what a faithful emitter can leak, not what it must report; emitter completeness is a separate property we do not prove. Third, the relation constrains nothing about the *values* an emitter writes and the theorem still holds—which matters because `begin` carries no postcondition on the values it returns. Definition 1 of the main paper's value conjunct only makes a witness harder to satisfy, so leaving values unconstrained strengthens the result rather than weakening it. Two witnesses pin non-vacuity in both directions: a window-free trace whose faithful emission is clean, and a windowed trace whose faithful emission does fire Definition 1 of the main paper. What remains trusted is the carrier correspondence—the theorem restates `a1_struct` over `int` and `Seq` where the deployed record uses `usize` and `Vec`, checked by inspection, the same link Theorem L2j discloses—and that no emitter in the artifact is itself proved faithful. `begin` snapshots the clock under a read handle; `commit` acquires the write handle, runs the same verified validation gate the deployed commit calls (Section 5.2 of the main paper), and either aborts or advances the clock, appends the versions, and appends the record—with the invariant as its postcondition. Because `acquire_write` ensures the predicate of the value it hands out and `release_write` requires it of the value it takes back, every state any thread observes, under any interleaving of begins and commits by any number of threads, satisfies the invariant; `trace_is_a1_free` states exactly this at a read handle, and a two-record witness shows the excluded window is satisfiable. Nothing is assumed about the lock: this artifact's trust base is Verus and `vstd` alone, and the three obligations of Section 5.2 of the main paper are not part of it. Two limits are stated rather than hidden: `vstd`'s lock specification does not verify deadlock-freedom (safety only, as throughout), and the values `begin` returns carry no specification, the theorem being about read and commit times. *This store is the deployed snapshot-isolation store.* The same file is compiled by plain cargo into the reference runtime (Section 5.2 of the main paper) as `si_concurrent.rs`, with proofs erased by the `verus!` macro, and `SnapshotIsolationStore`'s critical section is its lock: `parking_lot::Mutex` is gone from that store, the public API and both modes (default-SI's no-write bypass, which `tick` realizes; SSI's validation of every commit) are unchanged, and the integration tests pass unchanged, together with two new tests that drive the store from eight and from two contending OS threads. What remains outside the verified store is the interning of cell and value names to identifiers (an append-only bijection under its own lock, never held across the verified lock) and the abort counter, neither of which decides a commit. Two consequences are stated rather than hidden. The lock is a compare-and-swap spinlock (`vstd::rwlock`), adequate at the pilot's contention and not a general-purpose replacement for a parking lock. And the SI runtime's verification status changes shape: its A_1 freedom is now proved directly, as the structural form of Definition 1 of the main paper on the emitted trace under a verified lock, with no mutex assumption, whereas the refinements of Section 5.2 of the main paper link the *previous* representation of the store (the `BTreeMap` version chain) to the abstract SSI machine of Section 5.2 of the main paper; refining the concurrent store to that abstract machine, so that both routes meet, is the remaining proof step.

What that step will have to say. Before writing the proof we established that the theorem is true, and under which hypotheses, by exhaustive simulation. `python/l1_refinement_mock.py` transcribes both sides from their sources and checks every transition of the deployed store against the abstract machine over 506,880 transitions in four configurations. Begin, a successful commit and a validation abort simulate with *zero* counterexamples. Every failure is the store's guard refusal, and the guard is reachable only at the clock ceiling: the other disjunct, a snapshot whose read time exceeds the clock, occurs 0 times, because `begin_step` sets the read time to the clock and the clock is monotone. The refinement therefore holds under two hypotheses—the snapshot came from `begin_step` on this store, and *clock* $<$ `u64::MAX`—with the abstraction function that reads `store` and `last_write` off the flat version chain by taking, per cell, the greatest-time version, and `pending` from the outstanding snapshots.

Two limits are structural and no choice of abstraction function removes them, so the refinement theorem must state both. The concrete `Rec` carries no values—it is $\{agent, read_cells, read_time, write_cells, write_time\}$—while the abstract `OpRecord` carries the read and write values Definition 1 of the main paper ranges over, so the target is the value-erased projection of the abstract machine rather than its trace. And `pending` lives in the abstract state but not in `SiStore`, which hands a `Snapshot` to its caller and retains nothing, so the concrete side of the relation is the pair (`SiStore`, outstanding snapshots). This is a bounded check at one to two cells, two values, one to two agents and a clock ceiling of two or three—evidence that the proof is worth writing and a specification of what it must carry, not a substitute for it.

And whether the invariant is inductive. Simulation is necessary and not sufficient: a Verus refinement also needs an invariant *Inv* on the concrete state that is preserved from *any* *Inv*-state, not merely from reachable ones, since the verifier discharges the step obligation from *Inv* alone with no reachability to lean on. An invariant true of every reachable state but not inductive is true and unprovable by induction, and finding that out inside a proof attempt is expensive. `python/l1_invariant_mock.py` enumerates states satisfying *Inv* without regard to reachability and reports zero violations on both obligations: 89,759 *Inv*-states satisfy $Inv(c) \Rightarrow$ *all_invariants*$(\alpha(c))$, and 520,008 transitions out of arbitrary *Inv*-states preserve *Inv*.

It also settles which conjuncts carry weight, which matters because one of them is not in the source and cannot be. Three of `SiStore::inv`'s four are load-bearing: `trace_times` discharges `inv_clock_monotone` and `inv_record_writetime_le_clock`, `link` discharges `inv_last_write_dominates`, and ¬`a1_struct` discharges `inv_no_intervening_write`. `versions_le_clock` is not: nothing in `all_invariants` bounds `last_write` by the clock, and preservation survives its removal—it may serve other properties of the store, but it does no work here. The fourth requirement has no counterpart in `SiStore::inv` at all, because `SiStore` does not hold the outstanding snapshots: every such snapshot's read time must be at or below the clock, or `inv_pending_read_time_le_clock` fails under α. That is the second structural constraint above, now confirmed against the abstract invariants rather than inferred from the types, and it fixes the refinement invariant as a predicate on the pair rather than on the store.

The pessimistic store, the same way.

`lib_pess_concurrent.rs` (31 verified, 0 errors; no axiom, no `assume`, no `admit`, no `external_body`) does for the lock-based discipline what the store above does for validation, and is likewise the deployed code (`pess_concurrent.rs` in the reference runtime, with `pessimistic.rs` on top of it). Its lock predicate is the conjunction of the store's well-formedness with three facts about the lock table: holders are exclusive (no cell is held twice), every hold began at or before the clock, and *no agent other than the holder has written a held cell since the hold began*. The last is what makes window-freedom inductive rather than assumed: a committing operation is checked to have held each of its read cells since no later than its read time, so any other agent's write to such a cell is a version bounded by the hold's start and therefore outside the window, while the operation's own writes land at $clock+1$, after every record already committed. The excluded predicate is Definition 1 of the main paper without only its value-inequality conjunct—cross-agent kept, so it is implied by Definition 1 of the main paper and stronger to exclude—and a two-record witness shows it satisfiable. `begin`, `commit`, `release` and `tick` each preserve the predicate, so the lock's `release_write` obligation is discharged at every step. With this store no deployed *guarded* runtime locks with `parking_lot::Mutex`: what remains on it are the agent client's pending-operation cell and the vanilla store, which is the unguarded L_0 baseline and proves nothing. The three obligations of Section 5.2 of the main paper therefore cover no deployed guarded runtime; they are retained as the specification of what a lock must satisfy, for the abstract lift of the sequential refinements.

What the verified lock costs. Two properties changed with the primitive and are stated rather than measured away. `vstd`'s lock is a compare-and-swap spinlock, where the previous `parking_lot::Mutex` parked its waiters; both stores hold it only for the duration of a `begin` or `commit` step (microseconds—the cell holds of the pessimistic discipline span an agent's generation phase, but the *lock* does not), so at the pilot's contention the difference is not observable, while under heavy contention a waiting thread spins. And the hold table is a vector scanned linearly, so `begin` is $O(|cells| \cdot |holds|)$ and release rebuilds the table; correct and verified, but not the data structure a production runtime would choose.

APPENDIX I: PROBABILISTIC REFINEMENT OF A_1: THE CONSTRUCTION

Probabilistic refinement of A_1

The deterministic A_1 predicate of Section 3.1 of the main paper compares recorded values syntactically: a witness fires when $r_1.read_values[c] \neq r_2.write_values[c]$. Under stochastic generation that comparison and the operationally meaningful notion of *stale-read disagreement* diverge. Identical inputs induce identical output *distributions*—not identical outputs, since sampling alone separates two draws—while different inputs need not induce different distributions: a fresh re-read might leave the downstream behavior unchanged. This is the dominant objection to the determinism assumption, and we consider it sound. We treat it as a scope separation (Appendix I); this subsection records the specification-level construction and is explicit about the direction it does *not* establish.

Construction. We introduce an abstract spec function $disagreement_probability : PromptId \times Value \times Value \to \mathbb{N}_{[0,100]}$: the total-variation distance, in percent, between the distribution of the agent's downstream decision in prompt-context p having observed v_{stale} for cell c and its distribution had it observed v_{fresh}. The definition matters. Defined as the probability that two single runs differ—as an earlier version had it—the quantity is positive even at $v_{stale} = v_{fresh}$, and the bridge below is false. It is left uninterpreted at the specification level and estimated operationally by re-running the prompt under each value; Section 6.7 of the main paper carries out the population-level counterpart of that estimate for a categorical decision (the decision changes in 75% of flagged mismatches). The probabilistic A_1 predicate at threshold $\theta \geq 0$ leaves the structural conjuncts unchanged and replaces the value-mismatch conjunct with $disagreement_probability(p, v_{stale}, v_{fresh}) > \theta$. The construction is not mechanized in Verus. It uses one property, (b) $disagreement_probability(p, v, v) = 0$, which holds because the distance between a distribution and itself is zero.

What is and is not established. $A_1^{prob}(t,\theta) \Rightarrow A_1^{det}(t)$ for every $\theta \geq 0$: it is the contrapositive of (b), a one-line argument. The deterministic detector therefore flags every trace the probabilistic predicate flags, monotonically in θ. That is the *wrong* direction for a safety guarantee and the right one for a screen: the converse $A_1^{det}(t) \Rightarrow A_1^{prob}(t,\theta)$ fails, because traces exist where the recorded strings differ and the decision distribution does not. The deterministic detector is a sound *candidate-anomaly screen*—every probabilistically real event is in the audit set—and it over-reports by the factor Section 6.7 of the main paper measures.

Estimating the quantity from k re-prompts needs a concentration bound, and the relevant one is classical, not open: for the mean of k independent $[0,1]$-valued samples, Hoeffding's inequality [49] gives $\Pr(|\hat{p} - p| \geq \varepsilon) \leq 2e^{-2k\varepsilon^2}$. The budget an earlier version quoted—$k \geq 100$ at a ten-point margin—evaluates to $2e^{-2} \approx 0.27$, that is, 73% confidence and not the 95% a reader would assume; 95% at that margin needs $k \geq 185$. We do not mechanize the bound: Verus has no real-analysis development in which to state it [45], [58]. That is a limitation of the toolchain, and Appendix I states the resulting position.

Probabilistic case: a soundness-only screen

We do not claim a probabilistic refinement among this paper's contributions; framing one as a contribution would overstate what is achievable. The accurate position is a scope separation between two distinct questions, plus an open problem the current toolchain cannot close.

The deterministic detector answers a *per-trace* question exactly: did a stale read occur in *this* execution? It fires iff the trace contains a read whose observed value differs from a later committed write on the same cell, and for that question it is sound *and* complete—the characteristic function of the property (Section 5.1 of the main paper), with no false-negative concern, because the property is decidable from the trace. The *distributional* question—would re-executing the read–generate–write window diverge with probability exceeding θ?—is a counterfactual property of the prompt's output distribution, not of any single trace, and a non-firing detector on one trace does not certify it ($\neg$det may simply not have exercised the divergence on this run). We therefore use the deterministic detector as a per-execution *exact* detector and a per-prompt *candidate-anomaly screen*, and do not conflate the two.

Certifying the distributional property quantitatively—"the empirical estimate is within ε of the true rate with probability $1-\delta$ at $k = O((1/\varepsilon^2)\log(1/\delta))$ samples"—is Hoeffding's inequality [49],

a textbook result within reach of any proof assistant with a measure-theoretic probability library. What is missing is not the mathematics but its expression in Verus, which has no real-number probability theory; a count bound (the empirical count is at most the sample size) is all the current toolchain states, and it is not a concentration guarantee. We therefore make no probabilistic refinement claim, the mechanized development contains no probabilistic obligation and no probabilistic axiom, and an earlier version's description of the bound as an open problem is withdrawn: it is unmechanized here, not open. Carrying it into a Verus-style development—by IronFleet-style infrastructure [45] or a probabilistic program logic (Appendix R)—is formalization follow-up.

APPENDIX J: A_4 SPLIT-VIEW: THE VERUS FORMALIZATION

A_4 split-view formalization

Deferring A_4 (split-view under replication) would be a critical structural omission, because replication is central to deployed multi-agent infrastructure. A weak closure is available but worthless: "read-from-primary suppresses A_4" follows from the observation that A_4 requires two distinct replica ids while read-from-primary uses one—a structural tautology, since if only one replica is ever read, two replicas cannot be observed to disagree. This subsection gives an argument with content instead: a safety property of the replication *mechanism*, proved from an append-only invariant rather than read off the arity of the predicate.

Construction. The model (`verus-detector/src/lib_a4_split_view.rs`) represents the primary as an append-only, monotone-versioned log over a single cell (the per-cell argument carries split-view without loss, as A_4 is a per-cell predicate): the value committed at version k occupies log position $k-1$, the head version is the log length, and a write appends one entry and advances the version. A read records the replica it was served by, the version and value it observed, and its trace position; a primary read observes the head, while a secondary replicates the primary only up to some log index, and a *lagging* secondary serves an older index than the current head. The A_4 predicate is unchanged: a trace exhibits A_4 iff two reads of the same cell from *different* replicas observe *different* values.

Verified theorems (9 verified, 0 errors, 0 axioms, 0 `external_body`, 0 `assume`).

- **Append-only invariants (the content).** `lemma_state_monotone_len` proves the head version is monotone non-decreasing in trace order; `lemma_state_stable_eq_len` proves that two trace points with equal head version have *identical* logs—a write would have grown the length, so every intervening step was a read. Both are proved by induction on the trace and carry the argument the theorems below read off.
- **Theorem A_4-mono (`thm_primary_version_monotone`).** Primary read versions are monotone in trace order.
- **Theorem A_4-no-split (`thm_primary_no_split_at_version`).** Two primary reads of the cell that observe the *same* version observe the *same* value: no split view at a fixed committed version. This is the substantive safety statement, and it follows from append-only monotonicity, not from the predicate's arity.
- **Corollary (`cor_primary_diff_value_implies_diff_version`).** Two primary reads returning different values must be at different versions—the value-mismatch half of A_4 cannot occur among primary reads at one version.
- **Non-vacuity: the prevented phenomenon (`lemma_secondary_lag_admits_a4`).** A concrete trace in which a lagging secondary serves log index 0 (value NULL) while the primary sits at version 1 (value 10) produces a genuine A_4 witness. Both observed values are computed from the actual model state rather than asserted by hand, so the witness is faithful to the construction: this is exactly the split-view that monotone-primary pinning excludes, and the exclusion has a reason (the secondary served an older log index), not a definitional fiat.
- **Regime non-vacuity (`lemma_cross_version_primary_differs`).** Two primary reads legitimately differ in value (5 then 7) at distinct versions, so the no-split theorem does not hold vacuously by all primary reads being forced equal.

Scope. The theorem establishes that a single append-only, monotone-versioned primary admits no split view at a fixed version, and that the residual split-view arises precisely from secondary lag—which read-from-primary pinning removes. The model is single-cell and single-primary; eventually-consistent and CRDT-merge strategies remain out of scope, requiring an unbounded propagation horizon or merge-function algebra orthogonal to the argument here (Appendix R). As with the other model-level results, the A_4 model is not refined to executable code.

APPENDIX K: L_2–L_4: SCOPE AND COMPOSITION NOTES

Scope. The transitive-cascade case is not a residual: the causal-closure invariant *pred_closed* is proved an inductive invariant of the L_2 runtime model—a base case plus five per-step preservation lemmas (`lib_l2_safety.rs`: 33 verified, 0 errors, no `assume`, no `admit`)—so the one-level cascade is provably sufficient on every reachable state and the case is closed unconditionally. A constructive witness (Theorem L_2g, `lemma_no_cascade_admits_a3`) shows the excluded predicate is satisfiable, so the safety theorem is non-vacuous in the same sense as the L_4 witness of Section 5.4 of the main paper; reachability under a disabled cascade is not mechanized at the proof level, but it is exhibited operationally—the unguarded baseline of Section 6.3 of the main paper, which releases every operation's effects at commit, reaches the excluded footprint in 519–624 of 1000 seed-varied schedules at each depth, so the predicate the theorem excludes is not merely satisfiable in the abstract but reached by a runnable L_1-class baseline. We prefer this figure to the $1000/1000$ an earlier, structurally invariant harness produced: a baseline that fails in literally every scenario is evidence that the scenarios cannot discriminate, not that the predicate is easy to reach.

The correspondence to the cataloged A_3 is characterized precisely rather than asserted. L_2h prevents Definition 4 of the main paper exactly, on the same predicate the model and the executable runtime carry. For the flat-trace screen of Appendix G, L_2i discharges the value-provenance half (every committed read has a surviving producer of exactly its value), and the temporal half is closed by Theorem $L_2\mathrm{i}^+$ (`lemma_l2_reads_supported_temporal`): via a per-cell read-timestamp invariant *inv_read_temporal*, the surviving producer committed no later than the read observed it ($commit_time(w) \le read_at[t][c]$), a genuine consequence of the

monotone clock rather than an added assumption. The detector decides over an emitted Seq⟨*OpRecord*⟩ while the model is a state machine; that projection is discharged by Theorem L_2j below, and what the screen can and cannot see about Definition 4 of the main paper is R1–R5.

Scope. Theorem L_3a prevents A_6 *by exclusion*: saga well-formedness serializes external calls, so the model forbids the concurrent executions that alone could exhibit A_6, rather than ordering genuinely concurrent effects. We retain L_3a as the strict-serialization special case but no longer rest the A_6 claim on it. The concurrent case—A_6 prevention for genuinely concurrent external effects under an enforced commit order, which an earlier version left unmodeled—is now discharged by Theorem L_3g (`lib_l3_sequencer.rs`), which orders an arbitrary completion schedule into issuance order without excluding concurrency. The substantive contributions are therefore Theorem L_3g and Theorem L_3c (the orthogonal composition of the saga discipline with L_2's causal tracking); L_3a is definitional and is reported as such. The L_3 proof establishes the saga discipline as sufficient for A_6 prevention; the runtime engineering for real external-tool integrations (idempotency, timeout handling, partial-failure recovery) remains paper design, with the saga pattern itself [5], [39] as the operational realization template. The strong precondition on the append theorem requires all prior calls to have completed before the new call's issuance, which is operationally trivial under strict serialization but materializes the proof obligation explicitly. The L_4 step (adding $\neg A_2$) is treated next.

Composition and scope. A_2 prevention operates on the tool-registry domain, while L_3's saga discipline operates on the external-effect-call domain and L_2's causal tracking on the read/write/predecessor domain; the three domains are orthogonal, so a runtime applying all three prevents A_1,A_2,A_3, and A_6 simultaneously. This composition is now mechanized rather than argued: `lib_consistency_lattice.rs` defines the product state (L_2, L_3, L_4) over the three disjoint carriers and proves `lemma_lattice_jointly_safe`—that the conjunction of the three component invariants (*inv_l2* together with *inv_output_commit*, *satisfies_l3*, *inv_l4*) entails joint prevention of A_1 (image), A_3, A_6, and A_2—by invoking each component's safety theorem on its projection, with no `assume` and no `admit`. Three framing lemmas establish non-interference: a step on one carrier preserves the other two disciplines, since the predicates range over disjoint fields. The composition root verifies at 53 obligations (0 errors)—the three component modules re-verified together with five new composition lemmas. As with the lower levels, the proof establishes the discipline as sufficient for A_2 prevention at the model level; the runtime engineering for live registry hot-swapping (versioned signatures, snapshot lifecycle, validation-abort recovery) is the operational realization, not re-verified here. With L_4 closed, the strict chain $L_0 \subsetneq L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq L_4$ is mechanically realized end to end.

APPENDIX L: THE LATTICE DIAGRAM, AND THE SIXTEEN-POINT RUNTIME UNDER A WIDER VALUE DOMAIN

We chose this particular chain on operational grounds, not because it is canonical. The order $A_1 \to A_3 \to A_6 \to A_2$ in which anomalies are added reflects operational grounds. A_1 is the staleness failure of the basic read–generate–write window and the weakest non-trivial guarantee to establish first; we claim no frequency primacy for it—the corpus measurement finds its architectural precondition rare in that corpus (Appendix C), which scopes the anomaly to shared-mutable-state designs rather than showing it absent from them; the demonstrably in-the-wild failure nearest to it is the L_0 lost update (Appendix Q), which is not one of the four cataloged predicates, and bounded-staleness refinements in the sense of Bailis et al. [14] would yield intermediate sublevels we do not formalize. A_3 couples most directly with A_1 (read–write coupling on the value memory); A_6 completes the value-memory fragment (write–write atomicity within one operation); A_2 sits last because registry stability across the operation lifetime is the costliest single requirement. A reader with different priorities may prefer a chain admitting A_2 or A_6 first; all 24 are equally valid linear extensions, and the choice reflects expository convenience and the pilot's design, not mathematical primacy.

The eleven unnamed lattice points (the points not on the chosen chain — see Figure S1) correspond to model variants that combine anomalies in ways our operational pilot does not directly target. They are realized, and checked, by the parameterized runtime of Section 4.2 of the main paper; the chain is the subset that the deployed pilot exercises.

Visualization and the lattice table

Figure S1 displays $\mathscr{L}$ as a Hasse diagram, with the chosen L_n chain highlighted in bold. Table S7 tabulates which anomalies are admitted (✓) and prevented (×) at each named lattice point on the chosen chain.

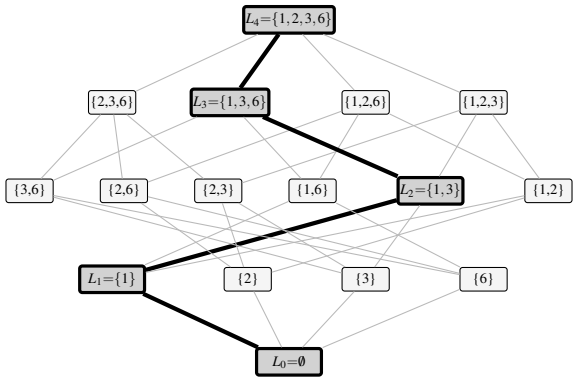


Fig. S1. Hasse diagram of the consistency lattice $\mathscr{L} = \langle 2^{\{A_1,A_2,A_3,A_6\}}, \subseteq \rangle$. Each node is the subset of anomalies excluded by that model, with set elements abbreviated by their anomaly indices (e.g., $\{1,3\}$ means $\{A_1,A_3\}$). Of the sixteen points and thirty-two cover edges, the chosen L_n chain (bold) is one of $4! = 24$ maximal chains. The remaining eleven unnamed points are realized by the parameterized runtime of Section 4.2 of the main paper.

The value domain, and what widening it changes. A single-element *Values* is the sharpest restriction in that configuration, and it is not cosmetic: `Guarded.tla` constrains a write to a cell of the write set to be non-NULL, so with one non-NULL value every write writes it, and an A_1 witness in the matrix can only be a read of the initial sentinel. A conflict between two *written* values is not expressible at that bound. We therefore re-ran the safety half at $\mathit{Values} = \{v_1, v_2\}$. All 32 safety runs remain SAFE, over 2,076,952 to 8,102,746 distinct states—a 6.1 to 8.8-fold growth, since the symmetry declaration covers agents and cells but not values—with no run exceeding a one-hour timeout or the 6 GB heap. These two-value figures (tracked in `v2_safe_cells.txt`) were measured on the model as it stood before G_3 became output commit—their

TABLE S7
Each named lattice point on the chosen chain admits (✓) or prevents (×) each of the four formalized anomalies. ADMITS column entries are verified by explicit TLC counter-example traces; PREVENTS column entries follow from the definitions of Section 4 of the main paper and are mechanically verified by the TLAPS coherence check (Section 5 of the main paper). The eleven unnamed lattice points are not tabulated; their admissions and preventions follow from their subset memberships and are checked individually in Section 4.2 of the main paper.

Level	A_1	A_2	A_3	A_6
L_0	✓	✓	✓	✓
L_1	×	✓	✓	✓
L_2	×	✓	×	✓
L_3	×	✓	×	×
L_4	×	×	×	×

6.1 to 8.8-fold growth is relative to that model's single-value counts (342,120–941,970)—and the two-value safety half has not been re-run since; for NOA3 it is a result about the superseded guard. The realizability claim is thus not an artifact of the single-value domain on the side where it could have been: the four disciplines still prevent exactly their avoidance sets when a second value exists for them to fail on. The 32 witness runs were not re-run, and do not need to be: widening *Values* only adds behaviours, so every witness trace at $Values = \{v_1\}$ remains a trace at $\{v_1, v_2\}$. What the wider domain does *not* do is make a new witness class reachable, and establishing that corrected an attribution we had made earlier.

Write NOA1NONNULLREAD for Definition 1 of the main paper restricted to the class a single-value domain cannot express: the stale read observed a value another agent had written, not the initial sentinel, and the superseding write carried a different real value. Checking it on the unguarded configuration across both parameters gives a complete answer:

	MaxOps=2	*MaxOps*=3
$Values=\{v_1\}$	unreach. (1,049,042)	unreach. (1,480,723)
$Values=\{v_1, v_2\}$	unreach. (9,227,002)	**reachable** (14,976)

Both conditions are necessary and neither suffices alone. A second value is required because with one non-NULL value the read and the superseding write cannot differ. Three operations are required because `Memory.tla` constrains a read to observe memory ($\neg AllowSkew \Rightarrow rv = memory$) and memory begins all-NULL, so the witness needs a seeding write, the superseding write, and the reader's own commit—three entries in a log that $MaxOps = 2$ caps at two. We had previously attributed the limitation to the value domain alone; the table shows that is wrong, and the sixteen-point matrix is bounded on the witness side by *MaxOps*. The safety side, which is what realizability asserts, is unaffected: all 32 safety runs hold at $Values = \{v_1, v_2\}$ as reported above.

Interaction, characterized. The matrix is also the interaction analysis. At the safety level the four disciplines do not interact: because the carriers are disjoint, enabling one guard neither creates nor removes any other anomaly—G_3's cascading retractions induce no stale generation, phantom tool, or reordering, and G_1's refusals induce no uncompensated cascade—and the check confirms this on every one of the 2^4 combinations rather than on the single three-way composition of Section 5.4 of the main paper. The interaction that does exist is at the cost level: a guard prevents by refusing a commit, a refused commit is a lost generation, and the lost generations are the abort-rate envelope of Section 6.6 of the main paper. Whether one guard's refusals raise another's—sequencing lengthens the window in which validation can fail—is a cost measurement, not a safety question, and is the empirical follow-up.

Scope. This is bounded model checking at small parameters, in the same sense as the witnesses of Section 3 of the main paper: witness traces embed upward into larger configurations; the safety runs are exhaustive within the bounds, not proofs. The Verus product of the L_2, L_3, and L_4 models (Section 5.4 of the main paper) mechanizes the all-enabled point; extending that product with the four switches is the proof-level follow-up. The 371-obligation headline of Section 5.2 of the main paper does not include these runs.

APPENDIX M: SYNTHETIC PILOT; COST FINDINGS 1–5 AND THE INFERENCE-COST SWEEP

Workload

The pilot uses seven synthetic but reproducible task scenarios designed to surface specific points in the lattice:

- **Concurrent stale-read.** Two agents read the same cell at the same clock tick, then commit divergent values in different orders. Designed to surface A_1.
- **Tool deprecation.** An agent plans a tool at `begin`; the tool is removed from the registry before `commit`. Designed to surface A_2.
- **Phantom-write cascade.** An agent reports having read a value of c that no committed write produced—a commit-log skew. Designed to surface the flat-trace residue of Appendix G; the synthetic traces carry no provenance or externalization fields, so Definition 4 of the main paper itself is not decidable on them.
- **Tool-effect reorder.** An agent issues writes in order $[x, y]$ but the store commits them in order $[y, x]$, simulating asynchronous tool execution. Designed to surface A_6.
- **Linear pipeline (clean).** Three agents update a shared progress cell in strict sequence. Control: no anomaly expected.
- **Fan-out merge.** A coordinator forks two workers writing disjoint cells, then merges. Branching control with no shared write target.
- **Single-agent control.** One agent makes three sequential updates. Single-agent control: no anomaly possible.

We executed 700 trace generations with scenarios drawn uniformly at random under a fixed seed, producing 700 JSON Lines traces. The first four scenarios are designed to surface one anomaly each; the remaining three are controls with no anomaly expected.

Witness classes, and the further runs.: The 100% rate in edit-review is structurally guaranteed by workload design, and the mechanism is worth naming because it is not the one the narrative of Section 1.1 of the main paper suggests. Classifying every committed Definition 1 of the main paper witness by what the stale read actually observed (`python/witness_classes.py`) splits them in two: a *cold-start* witness read the store's initial sentinel, so the reader reached the cell before anyone had written it, while a *supersession* witness read a value another agent wrote and had it overwritten inside its window. All 300 edit-review-vanilla witnesses across the three runs are cold-start, and 94.2% of the synthetic pilot's 485 witnesses are. The live study of Section 6.1 of the main paper is the opposite: all 292 of its witnesses are supersessions. The two halves of the empirical section therefore exercise different witness classes, and the synthetic rates should not be read as smaller-scale measurements of what the live study measures. The

same split bounds the formal side, though not for the reason we first recorded. At $|Values|=1$ the lattice matrix of Section 4.2 of the main paper can indeed express only the cold-start class, since a write to a cell in the write set is constrained non-NULL and one non-NULL value leaves a supersession with nothing to differ from. But widening the domain does not by itself reach the other class, and Section 4.2 of the main paper reports the check that shows why: the binding constraint is the operation budget, not the value domain. Plan-execute is 0/100; the 42% in triage is the only non-trivial intermediate measurement. Two further independent runs of the same protocol give edit-review 100/100 and 100/100, plan-execute 0/100 and 0/100, and triage 52/100 and 48/100; these are `20260913T0128Z` and the vanilla column of the Table 4 of the main paper run `20260913T0150Z`, both committed with their traces; pooled over the three runs, triage is $142/300$ [41.6, 53.2].

Table 4 of the main paper reports the level distribution. Three findings.

First, the edit-review rate of 100% confirms that under round-robin scheduling with both agents required to read the same cell before either writes, A_1 is *structurally guaranteed* rather than empirically observed. We retain this workload in the table not as a measurement of A_1 prevalence in the wild but as a calibration point: the rate at which the LLM follows the prescribed 2-call (read, write) pattern, with the residual 0% accounted for by sessions in which the model collapsed reads and writes into a single turn.

Second, the plan-execute rate of 0% ($0/300$ pooled over three runs, [0, 1.2]) confirms that strictly sequential workflows — where no cell is concurrently readable by two agents whose writes follow — suppress A_1; an earlier run reported a single firing that the regenerated traces do not reproduce.

Third, the triage rate of 42% (48% and 52% in two further runs; $142/300$ pooled) is the only measurement of the three that occupies a non-trivial intermediate regime. The 3-agent pipeline produces A_1 when the engineer's read of `ticket` interleaves with the triager's still-active read window, but not always; the variability comes from the LLM's tool-call ordering within each agent's turn, and the run-to-run spread at a fixed harness seed is the sampling variance of the model, which the seed does not pin.

Synthetic baseline comparison

To check whether the rates above are artifacts of an unguarded runtime, we re-ran the 700-trace synthetic pilot through three variants (vanilla; pessimistic locking, on-conflict ops silently dropped; snapshot isolation, on-conflict commits aborted) on the same scenarios, seed, and detector pipeline. Pessimistic locking drives both A_1 and A_2 to 0% at the cost of 285 dropped operations (40.7%); snapshot isolation drives A_1 to 0% but leaves A_2 at 21.1% (its scope is value memory, not the tool registry) at 145 aborts (20.7%); vanilla exhibits A_1 at 20.7% and A_2 at 21.1% with no aborts. Under these scripted scenarios both guarded runtimes pay ≥ 1 abort per anomaly prevented, and a sound-and-complete detector returning L_4 is machine-checkable evidence of an anomaly-free trace. This controlled-scale result is superseded by the LLM-scale baseline of Section 6.2 of the main paper, where each drop or abort corresponds to a re-issued inference call.

Observation: snapshot isolation's classical incompleteness manifests in the agent setting.: The SI runtime validates a committing operation against concurrent writers to its read set *at commit time*, but validation triggers only on a non-empty write set; a read-only operation ticks the clock and emits a no-write *OpRecord* without validation. This is the classical write-only SI behavior—read-only transactions in textbook SI are not validated [23], [38], and read-set serialization (SSI) closes the gap. We did not implement SSI in the pilot (a deliberate scope choice); the residuals are the fraction of sessions in which a read-only operation's decision reaches downstream agents without materializing as a validated write, not a general claim. Concretely, in each of the five edit-review residuals the reviewer's no-write commit at $t = 2$ follows the editor's write at $t = 1$ that landed inside the reviewer's window opened at $t = 0$; in the eight triage residuals the reporter writes the ticket twice and the triager or engineer, having read the first version, commits without writes against the second. The conceptual content—read-only operations that influence downstream agents need SSI-style validation—is identical to the database case; the operational regime in which it matters is what is new.

Pessimistic locking eliminates A_1 in every cell, by dropping the contested operation.: The pessimistic-locking runtime drops every conflicting operation silently (Appendix M), and every cell of Table 4 of the main paper shows zero A_1 instances—because in edit-review the reviewer's operation is gone in 99 of 100 sessions and in triage one of three agents' operations is gone in 95 of 100. The mechanism is stronger than SI's because the lock is acquired *before* the operation reads, and a failed acquisition drops the operation without invoking the LLM beyond the agent's planning phase; what it prevents is the stale commit, and what it costs is the second agent's contribution. It is the right mechanism only where a dropped contribution is acceptable and a wasted inference is not; where the contribution must survive, the L_1 discipline that keeps it is validation with retry (SSI, Section 5.2 of the main paper), whose cost is the re-spent generation of Section 6.6 of the main paper.

Cost of prevention.: Both disciplines drop a substantial fraction of operations under contention. Scored on the committed traces (`python/task_utility.py`), in edit-review pessimistic locking drops the reviewer's commit in 99% of sessions and default-SI in 95%, though the LLM inference still ran (SI aborts at validation, pessimistic fails acquisition—which could in principle short-circuit the call but does not in our implementation).

Prevention scored on the outcome, not on the anomaly rate. A discipline can drive the anomaly rate to zero by refusing every contested commit, and a token metric records that as cheap. We therefore score each cell on three quantities: *clean* (no Definition 1 of the main paper witness), *complete* (every role that commits in the unguarded cell of the same run also commits here—derived from the baseline rather than declared), and *both*. On edit-review the unguarded baseline is complete $100/100$ and clean $0/100$; pessimistic locking is clean $100/100$ and complete $1/100$; default-SI is clean $95/100$ and complete $5/100$, and those five are exactly the five that fired, so on this workload the reviewer's contribution reaches the log *if and only if* it is stale. No discipline, including the baseline, delivers more than one complete-and-clean session out of a hundred there. On triage the picture differs and is worth stating against our own recommendation: default-SI is complete-and-clean in $45/100$ while the unguarded baseline is in $52/100$, because default-SI drops the engineer in 47% of sessions. Only plan-execute, which aborts nothing, is unaffected. *Complete* is a structural proxy—it counts whether the baseline's set of contributors reappears, not whether a human would call the task done—and a session missing one role may still be useful output. With that caveat, a

prevention claim should be read against *both*, and the main paper's discussion's recommendation of snapshot isolation should be read as workload-dependent rather than general. A defensible cost claim requires per-session token counts from the client `usage` callbacks, which the original pilot did not capture (character-derived estimates underestimate cost by ignoring history growth and intermediate tool-call exchanges). We measure token-level cost at LLM scale in Section 6.6 of the main paper.

Finding 1: between sessions, snapshot isolation's token cost is not separable from the unguarded runtime's. The SSI mean sits within sampling noise of vanilla in every cell: CIs overlap on edit-review (3.11 vs 3.13), plan-execute (1.95 vs 1.93), and triage (3.13 vs 3.23). This is a statement about a per-session denominator, which credits a discipline for work it declined to do; Finding 6 re-prices the same run per sound committed operation, and the parity does not survive. An earlier version cited here a paired $\sim$8% overhead on plan-execute. That figure comes from the study of Appendix O, whose aborts are injected at a fixed probability; the real store never aborts on plan-execute, so the figure prices the injection and not the discipline, and we no longer cite it as a cost of snapshot isolation. The A_1-prevention rate of the pilot's default-SI store is 95%, 100%, and 92% respectively (Table 4 of the main paper); the SSI mode of the Rust runtime (Section 5.2 of the main paper), which validates read-only commits, is the mechanism the cost figures should be read against. The classical cost asymmetry that motivates serializable snapshot isolation in the database setting [23]—validation rather than locking keeps reads non-blocking—transfers to the agent setting in the form most favorable to deployment.

Finding 2: pessimistic locking incurs bounded overhead, concentrated in multi-stage contention. On edit-review the pessimistic overhead is statistically zero (CIs marginally overlap: 3.21–3.38 vs 3.11–3.16). On plan-execute the overhead is also zero (slightly cheaper than vanilla, within noise). On triage the overhead is real and statistically significant: \$5.21 m vs \$3.23 m, with CIs [4.90, 5.54] vs [2.93, 3.55] that do not overlap, an $\approx$62% increase. The pattern matches Appendix M: pessimistic locking adds cost where lock contention forces wait-and-retry across pipeline stages, and is essentially free in single-writer or low-overlap workloads.

Finding 3: the cost asymmetry in the agent setting is bounded at $\leq$1.6 $\times$ vanilla in the worst measured workload on gpt-4o, not the unbounded "orders of magnitude" that Section 2 of the main paper speculated. Our pre-measurement intuition—that a discarded LLM inference is "cripplingly expensive"—is empirically wrong for these workloads at this scale: worst-case overhead is bounded, typical-case near zero. Abort costs remain non-trivial in kind (Appendix M: pessimistic locking drops 40.7% of synthetic operations), but not order-of-magnitude in measured cost.

Cross-model validation on Claude Sonnet 4.5. To probe whether the gpt-4o findings reflect properties of the runtime model or of one specific model family, we replicated the full 900-session protocol against Anthropic's `claude-sonnet-4-5-20250929`, identical workloads and identical instrumentation. Table S2 (this appendix) reports the per-session cost. Claude pricing at the time of the run is \$3.00/M input tokens, \$15.00/M output tokens.

Rerun with operation records (run `20261014T0253Z-claude`, 2026-10-14). The replication above predates the operation recorder, so its sessions carry no A_1 verdict. The vanilla column was rerun on `claude-sonnet-4-5` with the recorder on, 100 sessions per workload, seed 42, and scored under Definition 1 by the same scorer as the gpt-4o cells: edit-review $100/100$ (the forced overlap, as on gpt-4o), plan-execute $0/100$ (the sequential control). Per session the vanilla cells cost 4,315 and 2,504 tokens (\$0.0162, \$0.0094; 8.3 s and 4.8 s) against 1,016 and 620 tokens on gpt-4o, in the same number of calls. The triage cell is *void*: in $100/100$ sessions the reporter wrote a ticket reading "no bug description provided" and the triager and engineer wrote nothing, so no operation of the pipeline could overlap a write; the same harness delivered the ticket text to gpt-4o's reporter (run `20260913T0150Z`: 17 distinct trace shapes, `priority` written in $28/100$ sessions). Whether the description was dropped by the provider adapter or declined by the model is not resolved by these records; the cell is reported as void, not as $0/100$. The guarded cells of this rerun did not execute (an import path broken by the repository migration) and remain listed as not run in the main paper. All 750 records pass the H0/H1 gates (`hypothesis_gates`: 0 overlaps, 0 read-after-commit).

Finding 4: SSI between-session cost-parity replicates across model families; pessimistic-overhead pattern is qualitatively consistent, quantitatively model-dependent. The strongest claim in this section—that snapshot isolation adds no cost overhead detectable between sessions vs vanilla— holds in both model families. All six SSI cells (three workloads $\times$ two providers) have means within sampling noise of their vanilla counterparts: CIs overlap on every cell. Pessimistic overhead retains the same shape on both providers (approximately zero on edit-review and plan-execute, real overhead concentrated in triage), but the magnitude of the triage overhead scales differently: gpt-4o exhibits a 62% overhead in triage-pessimistic, while Claude Sonnet 4.5 exhibits a 124% overhead in the same cell (\$27.73 m vs \$12.36 m vanilla, non-overlapping CIs). Wall-clock latency follows the same pattern: triage-pessimistic on Claude requires 21.4 s median wall-clock vs 10.5 s vanilla, a 2$\times$ latency penalty matching the cost penalty. The magnitude difference is not explainable from these data alone (candidate mechanisms: retry budgets, client-side tool serialization, tokenization density); the qualitative conclusion stands on both providers—bounded, $\leq$1.6$\times$ (gpt-4o) to $\leq$2.3$\times$ (Claude), never order-of-magnitude.

Finding 5: SSI costs one re-spent generation per abort, and a re-spent generation costs $1.074 \times [1.052, 1.096]$ a first-attempt one. To test the discipline where its abort-and-retry machinery engages, we ran a paired high-contention sweep (Figure S2): contention varied two independent ways—fan-in $W \in \{2,4,8,16\}$ at one cell, and cell count $C \in \{2,4,8,16,32\}$ at W=8—over 155 paired sessions on gpt-4o-mini at realized abort rates 0.09–0.93. Token cost is counted exactly (including generations re-spent on aborts), and the harness re-verifies per session that the baseline exhibits A_1 while the guarded disciplines do not—scored with the superstep form of the predicate (Appendix C), since each round of the sweep is a single superstep whose commits land at one tick—so the measured quantity is the cost of *working* prevention.

What in this sweep is arithmetic. Regressing token overhead on the realized abort rate returns $+0.1\% + 108\% \times$ abort_rate at $R^2 = 0.94$, and that regression is an accounting identity rather than a result. The committed sessions show why: in all 155 SSI sessions the harness records $\mathit{generations} = W \cdot \mathit{depth} + \mathit{aborts}$, and in all 155 vanilla and 155 pessimistic sessions $\mathit{generations} = W \cdot \mathit{depth}$. The realized abort rate $\mathit{aborts}/(W \cdot \mathit{depth})$ is therefore exactly *extra generations per baseline generation*, while token overhead is extra

tokens per baseline token; regressing the second on the first recovers one quantity only—the price of a retried generation relative to a first-attempt one—and the R^2 measures how tightly tokens track generations, not how well a law fits. The collapse of the two contention mechanisms onto a common line is forced for the same reason: any mechanism that produces aborts lands there. `python/fig_cost_envelope.py --stats-only` prints the identity check and every figure below.

What the sweep establishes. Four things the identity does not force. *(i)* The fitted intercept is indistinguishable from zero ($+0.05\%$, 95% CI $[-1.5, +1.7]$), and the eight sessions that aborted not at all cost $+0.56\%$ in the mean: SSI carries no fixed overhead, only a per-abort one. *(ii)* The per-abort price does not steepen with contention—below abort rate 0.5 the slope is $110.7\%\,[99.8, 121.4]$ and above it $106.5\%\,[97.5, 115.2]$—so aborts do not compound, which a discipline that re-validated a growing read set on every retry would not satisfy. *(iii)* The two contention mechanisms agree on the per-abort price in absolute tokens ($1.077\,[1.05, 1.11]$ for the fan-in sweep, $1.071\,[1.04, 1.10]$ for the cell-count sweep), so the retry price is a property of the discipline rather than of how contention is induced. *(iv)* A retried generation costs $1.074\times$ a first-attempt one (95% CI $[1.052, 1.096]$, bootstrap over the 147 sessions with at least one abort): re-prompting after a refused commit is measurably dearer than the original attempt, because the retry carries the conflict into its prompt. That ratio is the entire empirical content of the 108% above, and Figure S2B shows it directly, against the null in which a retry costs exactly one generation.

The practical envelope follows from *(i)–(iv)* rather than from the regression: overhead is zero when nothing aborts and 1.074 generations per abort thereafter, so it reaches 15% at an abort rate of 0.14—which is $0.15/1.074$, arithmetic on the measured retry price and not a fitted breakpoint. The lowest cell measures $+9.9\%\,[5.6, 14.5]$ at abort rate 0.09, so that regime is measured rather than extrapolated, and what it measures is low overhead, not zero. We do not establish that deployed workloads occupy it (the corpus audit of Appendix C is consistent with it but is not an abort-rate measurement); the envelope instead makes the question auditable per deployment. Pessimistic locking is the complementary profile: token-neutral between sessions ($\pm 1\%$) but serializing critical sections, so its cost is real yet invisible to a token metric (wall-clock figures here are *composed* from per-call latencies, not measured under a live concurrent runtime, and grow several-fold at 16-way).

What this section cannot establish. The measurement covers gpt-4o and Claude Sonnet 4.5 on these three workloads. Cost behavior on Gemini or other providers, and on workloads with materially larger read-sets, is not measured; the high-contention sweep of Finding 5 covers fan-in up to 16 agents and multi-round pipelines, but only on gpt-4o-mini. The measured cost asymmetry is consistent with the design intuition that SSI offers a near-optimal cost-correctness trade-off in low-to-moderate contention regimes and pessimistic locking is preferable only under workload-specific contention patterns; the two-provider replication strengthens this claim qualitatively but does not generalize to open-weights models or to model families optimized for low-latency tool use. Three-plus provider replication including local open-weights inference (Llama, Mistral) and higher-contention workload exploration are identified as principal empirical follow-up (Section 8 of the main paper).

TABLE S8
Operational disagreement probability p_{op}: fraction of flagged events that change the agent's decision. The controlled task uses an exact categorical comparator; pilot-trace free-text outcomes are LLM-judged at decision level (a softer signal). All pilot events are read-before-write ($v_{\text{stale}} = \text{NULL}$). Only the reviewer row is scored on Definition 1 of the main paper firings; the engineer and editor rows are scored on later-rewrite events (the rewrite landed after the reader's commit), which the corrected predicate does not classify as A_1.

Setting	Role / decision	p_{op}	95% CI
Controlled	triage priority (categorical)	75.0%	[68.4, 80.8]
Pilot (A_1)	reviewer (assess doc)	100%	[91.8, 100]
Pilot (later rw.)	engineer (resolve, judged)	54.5%	[36.4, 71.9]
Pilot (later rw.)	editor (author, judged)	0%	[0, 3.6]

APPENDIX N: OPERATIONAL MATERIALITY: TABLE, SCOPE AND THE ADVISORY LAYER

Scope.: The controlled task is a proxy with an exact comparator; deployed free-text outcomes are judged by an LLM of the same family under a decision-level rubric and are reported as a softer, corroborating signal. On a blind sample of 60 judged firings—the coder shown only the two free-text outputs, not the model's verdict—manual coding agreed with the LLM judge on 59 of 60 (98%; Cohen's $\kappa = 0.96$ between judge and human, approximate 95% CI $[0.89, 1.00]$), so the judge tracks a human reading of decision-level divergence rather than a lexical artifact. We state the sample's composition rather than let it be inferred: those 60 rows are 35 editor, 10 engineer and 15 reviewer events, so 45 of them belong to the later-rewrite class this subsection withdraws, and only 15 of the reviewer's 43 genuine firings were blind coded. The agreement figure is therefore evidence that the judge tracks human decision-level reading in general, not a validation restricted to the 43; re-coding those 43 is outstanding. All pilot firings are read-before-write, so value-staleness is exercised only by the controlled experiment. The editor and engineer results reflect these workloads, in which the producer does not fully consume the read; they are not a claim that producer roles are universally immune. The triager cell had only three qualifying firings, too few to estimate a rate, and is omitted from Table S8.

Design consequence: an advisory materiality layer over the verified core.: The controlled result—one flagged mismatch in four did not change the decision—bears directly on the objection that a boolean detector firing can be operationally meaningless. The verified core remains the boolean predicate of Section 5.1 of the main paper; we do not weaken its sound-and-complete guarantee. A deployment can wrap it in an *advisory* pair $(\textit{detected}, \hat{p}_{\text{op}})$, with $\hat{p}_{\text{op}}$ estimated per prompt by the counterfactual re-prompt of this section, so that an abort policy gates on $\hat{p}_{\text{op}} > \theta$ rather than on the raw firing. An earlier version proposed estimating $\hat{p}_{\text{op}}$ from the firing agent's *role*; the role-dependence that rested on is withdrawn above, and the knob is withdrawn with it. The layer is deliberately unverified and outside the trust base: it changes which firings a runtime *acts* on, not which the verified detector *reports*.

APPENDIX O: THE INJECTED-ABORT WALL-CLOCK STUDY

Within-model wall-clock study (injected aborts)

The token-instrumented cost study of Section 6.6 of the main paper reports relative overhead within each provider but invites the objection that any comparison *across* providers conflates runtime overhead with tokenization, pricing, and serialization differences,

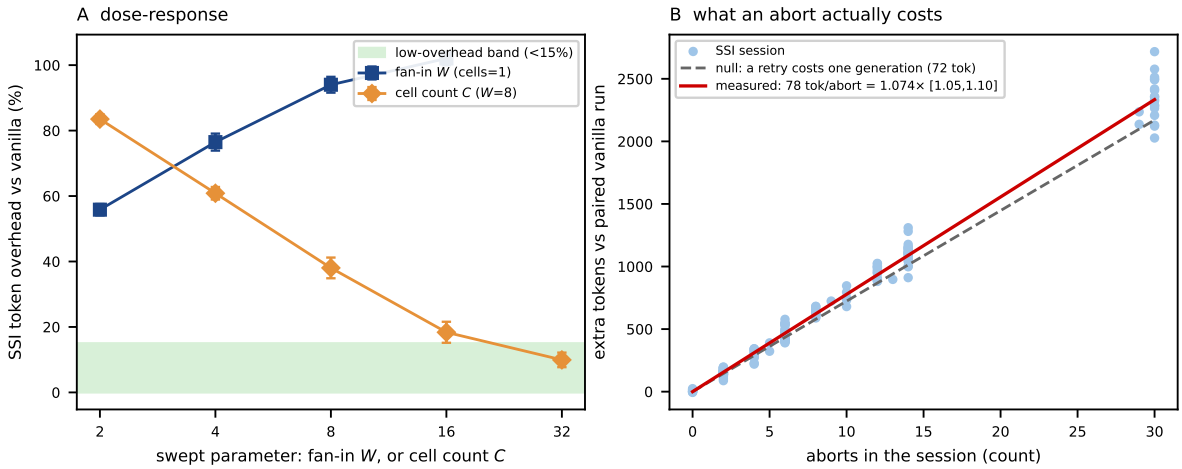


Fig. S2. The high-contention sweep (gpt-4o-mini, 155 paired sessions), separated into what the accounting identity forces and what the data measures. **A**: mean SSI token overhead against the parameter we *set*—fan-in W at one cell (squares) and cell count C at $W=8$ (diamonds), error bars ± 1 s.e.—rather than against the realized abort rate, which is another outcome of the same runs. The two knobs move overhead in opposite directions and the sweep crosses the low-overhead band ($<15\%$; a descriptive plotting threshold—the pre-declared equivalence margin of Appendix O is the stricter $\pm 10\%$). **B**: extra tokens against aborts in absolute units. The dashed line is the null in which a retried generation costs the same as a first-attempt one (72 tokens); the fitted line is the measurement, 78 tokens per abort, a ratio of $1.074\,[1.052, 1.096]$. The gap between the lines is the only quantity the former 108% slope carried. Pessimistic locking is token-neutral in between-session means but incurs the wall-clock cost reported in the text. Regenerate with `python/fig_cost_envelope.py`.

and that billed tokens are not the same quantity as wasted wall-clock computation. This subsection reports the instrument we identified as the clean resolution: a study that holds the model fixed (gpt-4o), varies only the runtime strategy, and measures *wall-clock seconds* rather than billed tokens.

Method. We ran 270 sessions (`python/wallclock_cost_study.py`): three workloads (edit-review, plan-execute, triage) $\times$ three strategies (vanilla, pessimistic with a 20% per-write abort probability, SSI with a 5% commit-time abort probability) $\times$ 30 sessions, on gpt-4o (and replicated on open-weights Llama-3.2 below) with deterministic mock tools so that the measured wall-clock isolates LLM-inference and abort-and-retry time from external-service latency. We report bootstrap 95% confidence intervals (5000 resamples) on the per-cell mean wall-clock.

The aborts in this study are injected. Each strategy is simulated by wrapping the workload's tools so that a write fails with the stated fixed probability and is retried, independently of contention; neither store of Section 5.2 of the main paper runs. The study therefore measures what a retry costs, in wall-clock and in tokens, at abort rates we chose. It does not measure the abort rate either discipline produces, and no comparison between the two strategies' overheads can be read from it, since the 20% and 5% are inputs.

Finding. The wall-clock overhead of both guarded strategies over vanilla is small and *not statistically separable from API-latency noise at $n = 30$ in any of the three workloads*: every pessimistic and SSI confidence interval overlaps the corresponding vanilla interval. The directional pattern is nonetheless the expected one. The overhead is largest where a session makes the most writes and so draws the most injected aborts—triage, in which the 20% injection produced 14 aborts across 30 sessions versus 6–7 in the other workloads—and the triage pessimistic interval ([4.82, 5.49]) sits almost entirely above the vanilla interval ([4.57, 5.07]), a $+6.6\%$ point estimate. The within-cell token overhead mirrors this: $+5.6\%$ on triage, $+3.4\%$ on edit-review, -1.4% on plan-execute. The two apparent inversions where a guarded strategy is point-faster than vanilla (edit-review pessimistic and SSI) are artifacts of API-latency variance: vanilla drew several high-latency outliers, and the strategies do near-identical compute when realized abort counts are low.

What this resolves, and what it does not. The within-model design removes the cross-provider confound by construction: every comparison in Table S5 (this appendix) holds gpt-4o fixed, so tokenization, pricing, and serialization differences cannot enter. And measuring wall-clock rather than billed tokens addresses the "tokens are not wasted computation" objection for the quantity measured: the real-time price of retry at the injected rates is single-digit-percent and at the noise floor, smaller than the billed-token figures, because the additional tokens of a retried operation are largely cached-prompt context that is processed quickly relative to a fresh generation. The conclusion is correspondingly narrow: at injected abort rates of up to 0.47 per session, retry imposes no wall-clock cost separable from inference noise at $n = 30$. Whether the stores' *real* abort rates on these workloads fall in that range is answered by Table 3 and Table 4 of the main paper, not here, and on edit-review they do not: the reviewer's commit is refused in 95–99% of sessions.

Open-weights replication (Llama-3.2). We re-ran the identical 270-session protocol on open-weights Llama-3.2 served locally through Ollama, where dollar cost is negligible and wall-clock is the only meaningful unit (Table S6 (this appendix)). Per-session wall-clock is an order of magnitude larger than on gpt-4o (local inference at 24–38 s versus 5–9 s), and against that latency the abort-and-retry overhead of both guarded strategies is invisible: every pessimistic and SSI interval overlaps the corresponding vanilla interval, with point ratios in $[0.82, 1.04]\times$ and SSI point-faster than vanilla in all three workloads—an artifact of the large per-call latency variance, not a real speed-up. The open-weights result extends the noise-floor conclusion to a third model family: at the injected abort rates, retry imposes no wall-clock cost separable from inference noise on local Llama-3.2 either.

Scope. This is a two-model-family ($n = 30$ per cell each) study

TABLE S9
The L_2 measurement on 1,000 seeded schedules per depth (`examples/l2_measure.rs`; main paper, Section 6.3). Every figure is decided by the schedule and the store; none is an estimate. Exposed effects: count (scenarios).

	Arm	depth 2	depth 3	depth 5
Definition 4 (of 1000)	verified	0	0	0
	unguarded	519	564	624
	cascade	519	563	624
Surviving-dependent (of 1000)	cascade	0	0	0
Exposed effects	verified	402 (199)	547 (199)	549 (199)
	unguarded / cascade	1,538 (771)	2,059 (771)	2,010 (771)
Relabeled after release	cascade	767	1,288	1,239
Release refused at least once	verified	748	1,255	1,272
of which released on approval	verified	184	315	382
Committed-and-unaborted released	verified	2,870/2,870	3,494/3,494	5,545/5,545

with mock deterministic tools; its aborts are injected rather than produced by the stores, and it does not capture real-tool latency, and at $n = 30$ the API-noise floor exceeds the overhead signal, so the study bounds the overhead from above rather than resolving its exact magnitude. A higher-contention workload, larger n, or real external tools would be required to separate the triage signal from noise. The `wallclock_results.json` output is in the artifact bundle.

Paired re-analysis and power. These are the injected-abort sessions above, so every overhead below is the price of the injected retries, not of a store. Because the same per-session seed was used under each strategy, a paired analysis (matched on ⟨workload, seed⟩) removes between-session variance and yields an explicit MDE at α=0.05, power 0.80. On *token* cost it separates two small overheads from zero—SSI on plan-execute (+91 tokens, ∼8%, $[+32, +150]$, p=0.007) and pessimistic on triage (+41 tokens, ∼5.5%, p=0.04, nominal only: not surviving Holm correction across the six tests)—while pessimistic-on-plan-execute and SSI-on-triage are genuine nulls (MDE ≈ 54–87 tokens, below the 10% effect of interest), established as *equivalent* by a two-one-sided test against the pre-declared ±10% margin. SSI-on-plan-execute is *not* equivalent (90% CI $[+40, +140]$ vs. a 111-token margin: up to ∼12.6%), and edit-review is underpowered (MDE ≈ 290 tokens). On *wall-clock* time the paired MDEs (0.7–3.4 s) exceed the candidate effects and the lone nominal cell is in the cost-*reducing* direction (residual latency confounding), so we claim neither overhead nor equivalence there. The paired analysis thus quantifies the earlier statement: the token price of the injected retries is ≤ 8% where separable, and their wall-clock price is below the resolution of an n=30 design.

APPENDIX P: L_2 MEASUREMENT, TOOLNODE AND COST-METRIC NOTES

What erasure costs the caller, and what the runtime now does about it. Under plain `cargo` every `requires` clause of the exec API disappears, so a caller that breaks one corrupts the runtime silently. An earlier version of this section stated that limitation and discharged the obligations in the measurement driver. The runtime now discharges them itself: every mutating entry point checks its own precondition and refuses instead of assuming. `begin` returns `Option`; `write`, `read`, `commit`, `abort` and `externalize` return whether they acted; `commit` decides `commit_valid` with the verified `can_commit` rather than trusting the caller; and each contract states both that the return value equals the exact guard and that a refusal leaves the abstract state unchanged. The only remaining precondition is `wf()`, the type invariant the API itself maintains. What is still trusted is stated in the trust base: the runtime's fields are `pub`, so a caller that writes them directly is outside what is verified.

Three properties of those figures are worth stating rather than leaving to be checked. The baselines fail *often but not always*: a baseline that fails in every scenario would make the comparison a tautology, and the earlier 1000/1000 was exactly that. The gate refuses in every cell, so the freshness check is exercised rather than inert—under the previous schedule it refused nothing at all, and a correct gate and a gate with no check were therefore indistinguishable in the measurement. And a differential regression test pins the difference directly: on a read-modify-write schedule the superseded carve-out gate accepts a commit that `commit_valid` forbids, so the defect cannot be reintroduced silently.

We state the shape of that evidence rather than let the pooled figure carry it. `Llama-3.2` supplies 62 of the 66 retractions; the two API models contribute four events between them, so a per-model statement for them would rest on $n = 3$ and $n = 1$. We make the pooled statement and no per-model one, and raising the API models' n would mean engineering the supervisor prompt toward retraction—which would make the rate an artifact of our design, the objection Section 6.9 of the main paper already concedes elsewhere. The prevention is structural: the executor depends on the planner because it read the plan cell, and output commit holds the executor's effect until the plan's review is decided, so a retracted plan takes its held dependent with it before anything is out; the 0% holds by construction of the verified discipline regardless of model. The live runs therefore do not claim the model's reasoning *causes* A_3; they demonstrate the verified discipline operating under real, divergent agent behavior (a 0.5–31.0% spread of supervisor judgment) at real cost and liveness, closing the gap between the model-level proof and a running multi-agent loop.

The 1,000-session haiku replay (2026-10-14). To put a per-model number on the family that retracts least, the live driver (`crates/runtime/l2-live`, seed 20261013, LLM supervisor) ran 1,000 further `claude-haiku-4-5` sessions through the verified and the unguarded arm: 0/1,000 retractions; hold of kept effects median 687 ms, p90 789 ms; both arms released all 1,000 executor effects, and A_3 could not arise on either. Over 1,200 haiku sessions the supervisor has retracted once; the hold is the supervisor call itself (median 687 ms), the whole price of output commit on a workload whose reviews are never adverse, and the hazard L_2 removes is on this evidence a property of the backbone's willingness to retract (62/200 on Llama-3.2), not of the runtime (`results/live_l2_haiku1000/`: per-session JSON, effect logs, `summary.json`).

Mechanized refinement of the executable runtime. The measured bytes are themselves the verified ones: `lib_l2_exec.rs` implements the L_2 protocol in Verus *exec mode* and is mechanically verified to refine the abstract model. Each state transition (`new`, `begin`, `read`, `write`, `commit`, `abort`, `externalize`) carries a postcondition equating `self.view()` with $\texttt{step}_X$`(old.view(),...)` when the call acts and with `old.view()` when it refuses, where `view()` maps the executable, `u64`-keyed runtime state into the model's `RuntimeState`, and each preserves the

well-formedness predicate $wf \equiv inv_l2 \wedge inv_output_commit \wedge fresh_ok \wedge keys_contiguous$. The model's A_3-freedom theorem (`lemma_l2_reachable_no_a3`) therefore transfers to the executable runtime as a capstone (`a3_free`): every well-formed runtime state is A_3-free, by construction of the transitions that produce it. Crucially, the interning of heap identifiers as `u64` keys is *natively injective*—the `u64` → `int` embedding via `as int` requires no axiom—so the exec-to-model refinement adds *zero author-introduced* entries to the trust base. (The exec state is `HashMap`-backed and so relies on vstd's standard `group_hash_axioms`, trusted with the Verus standard library; we add no axiom of our own.) This is a stronger footing than the four spec–runtime refinements of Section 5.2 of the main paper, which share two structural axioms. The file re-includes the L_2 model verbatim and verifies at 73 obligations end to end (0 errors, 0 `assume`, 0 `admit`, 0 added axioms).

The live replay now runs on the same artifact. It was produced by the superseded twin in an earlier version and is re-run here through `src/l2_exec.rs`, gated by the verified `can_commit`, over sessions that are committed with the artifact. What remains weaker about it is not the runtime but the sample: two of the three model families retract too rarely under a neutral supervisor prompt to support a per-model claim, and we say so where the figure is reported. Second, the exec runtimes are sequential objects: a multi-threaded caller must serialize its calls with a lock of its own choosing, which is outside what is verified (the L_0/L_1 stores, by contrast, carry their verified lock inside, Section 5.2 of the main paper). Third, the workload is a synthetic cascade against our own baselines, so the measurement shows the discipline *executes and prevents A_3 as proved*, not that uncompensated cascades occur at this rate in the wild (the prevalence question of Appendix C is separate). The same two-artifact closure extends to L_3 and L_4: `lib_l3_exec.rs` (7) verifies an executable sequencer whose invariant pins emitted order to issuance order (A_6-freedom by invariant), and `lib_l4_exec.rs` (11) verifies a dispatcher that returns a binding exactly when dispatching against the *live* registry is not A_2 and refuses otherwise; both are self-contained (zero axioms). L_3's measurement runs a dependency-free twin under seed-varied schedules, with completeness guards confirming prevention is not by dropping effects; L_4's runs the verified dispatcher itself.

What the baselines measure, and what they do not. An earlier form of both twins fixed its own baseline. The A_6 generator reshuffled inside a rejection loop until the permutation was not the identity—the one schedule under which A_6 cannot fire—so the unsequenced baseline reported $1000/1000$ for every width and seed by construction, and the test asserted that figure. The A_2 generator rewrote the planned tool's signature unconditionally after its churn loop, so its baseline was likewise $1000/1000$ by construction. Both constructions are removed. The A_6 baseline can now be checked against a closed form: a uniform random permutation is the identity with probability $1/w!$, so it is $487/1000$, $955/1000$ and $1000/1000$ at widths 2, 4 and 8 against a predicted $1 - 1/w!$ of 0.500, 0.958 and 0.99998, with prevention $0/1000$ in every cell.

Grades of evidence. L_2 and L_4 are verified and measured on the bytes Verus verified, and L_2 additionally on 600 recorded live sessions (above); L_3 remains verified at the model level and measured on a separate twin, which the trust base records. For A_6, Section 6.4 of the main paper adds an intermediate grade of evidence—the anomaly and its L_3 prevention demonstrated against a released framework on real model-emitted call orders, with tool effects modeled.

The intended order is genuine, and the reorder is latency-driven. We reproduced the defect on the released `ToolNode` (`langgraph` 1.2.5) with an order-dependent pipeline of five tools (`create_database` → `run_migrations` → `seed_reference_data` → `build_search_index` → `enable_live_traffic`) under a natural prompt that states the task and the tools with no instruction to parallelize. The model emits the calls in canonical order within a single turn; that emitted sequence is the issuance order *io* of Definition 5 of the main paper, not an order we imposed. Two controls isolate the mechanism. (i) Under *equal* per-tool latency the committed order equals issuance in every run—a uniform-latency control yielding 0 reorderings—so the reorder is a function of latency *differences*, which real tools always exhibit (the stateful UI actions of issue #861; a cache read beside an external write). (ii) In every trace the committed order is the latency argsort, $co = \text{argsort}(\text{latency})$, checkable per run. With the fastest call committing first, the harness routed live traffic to a database it had not yet created.

Whether a model triggers it varies sharply—which is the argument for L_3, not against it. The trigger is whether a backbone batches order-dependent calls into one turn, and that behavior is model- and version-dependent, not something a deployment controls. Under the natural prompt, `gpt-4o-mini` batched in $30/30$ turns and reordered in 22 (including all 18 turns in which it batched the full five-step pipeline); `llama-3.2` batched in $3/30$ (reordering 2); and `claude-haiku-4-5` batched in $0/30$—across 60 turns under two prompt conditions it issued one call per turn and never exhibited A_6. A backbone that batches walks into the defect; one that sequences sidesteps it. The point is not that one model is wrong, but that effect-order correctness would otherwise rest on a property no specification pins down: the same task yields A_6 on one backbone and not another. The reordering rate scales with batch width (`gpt-4o-mini`: $4/12$ at width two, $18/18$ at width five), confirming that *co* tracks completion rather than issuance. We report these per model and deliberately do not pool them into a single headline rate; the rate is a property of the workload and the backbone, not of the framework alone.

The anomaly is invisible in the transcript. In all batched turns the returned `ToolMessage` list preserved issuance order ($33/33$), because `gather` orders *results* by input position while the *effects* have already committed out of order. The conversation history a developer would inspect shows the correct order; the external state does not. A_6 is thus undetectable by reading the artifact most operators check, which is why a runtime-level discipline—rather than a prompt instruction or a transcript audit—is the appropriate remedy.

Two clocks and one anomaly. Reported as three figures—42, 0, and a summary grounded in a superseded plan in $300/300$ sessions—this invites the objection that A_1 is an artifact of the clock rather than an anomaly. It is not, and the relation is formal. `tla/MC_ClockSensitivity.tla` carries both clocks on one record and adds a third predicate, A1SEMANTIC, that mentions no completion order at all: a different agent superseded the reader's value no later than the reader's own barrier. `tla/check_clocks.sh` settles six claims by TLC over 747,361 states, each carrying exactly the hypotheses it needs.

Both clock-bearing predicates imply the semantic one. A1BARRIER ⇒ A1SEMANTIC needs no hypothesis;

A1TEMPORAL ⇒ A1SEMANTIC needs exactly one, that the completion clock refines the barrier order—a node body cannot return before a barrier preceding it has closed—and dropping it makes the implication false, which the check confirms rather than assumes. So $42/100$ is a *lower bound* on $300/300$, not a rival measurement. The converse fails, and TLC's counterexample is the case the harness actually produces: two bodies returning in the same completion slot, where Definition 1 of the main paper's strict window cannot see a supersession that happened. That gap is the distance between 42 and 100.

The $0/100$ deserves the sharpest statement of the three, because it is *definitional*. When every write lands in one barrier, A1BARRIER's window requires $write_barrier_j < write_barrier_i$, which one barrier makes impossible: the barrier clock reports zero whatever the agents did. The completion clock is not blind there, and the check shows that too. A zero under the barrier clock is therefore evidence about the clock and not about the execution, and we do not offer it as prevention.

Cost-metric limitations and silent failures. The cost analysis of Section 6.6 of the main paper reports per-session token counts, the metric the LLM providers expose through usage callbacks. This is an incomplete proxy for the cost-asymmetry argument in two specific ways. First, an aborted commit after a 30-second generation is *wall-clock waste* that is not fully captured by the token count, since the inference phase has already consumed compute regardless of whether the commit succeeds. A within-model wall-clock measurement (same model, different runtime strategies) isolates the runtime-overhead question; we report one in Appendix O, but its aborts are injected at fixed probabilities rather than produced by the stores, so it prices a retry and does not bound either discipline's wall-clock overhead; a wall-clock study on the real stores is outstanding. The token-cost finding should accordingly be read as "no order-of-magnitude token-cost asymmetry" rather than "no runtime-cost asymmetry." Second, snapshot isolation's residual A_1 rates of 5% in edit-review and 8% in triage (Table 4 of the main paper) reflect read-only operations that bypass SI validation; these are *silent*—they succeed from the runtime's perspective and consume tokens normally—but produce stale outputs. The between-session cost-parity finding therefore cannot be read as "SI is operationally equivalent to vanilla"; it is "SI's per-session token cost overlaps with vanilla's, modulo a silent residual anomaly rate that does not appear in the cost metric."

Cross-model cost comparison is confounded. Pricing, tokenization density, tool-call serialization, and retry budgets differ between gpt-4o and Claude, so the magnitude difference in pessimistic overhead ($1.6\times$ vs. $2.3\times$) cannot be attributed to runtime strategy alone; we make only the within-model claims of Section 6.6 of the main paper.

The operational model itself — single shared key-value store, discrete monotonic clock, in-process agents — abstracts away network partitioning, replica divergence, and asynchronous tool execution, whose consequences are discussed in Section 8 of the main paper.

Concurrency-model granularity. The abstract-machine safety theorems and the four refinements hold for sequential interleavings of atomic begin/commit (and abort) transitions. For the two deployed guarded stores the lift to multi-threaded execution is proved, not assumed: every state any thread observes satisfies the store invariant under `vstd`'s verified reader–writer lock (Section 5.2 of the main paper). A mutex assumption survives in two places only: the abstract lift of the sequential refinements, bounded-checked under RC11 with GenMC at small agent counts, and the L_2–L_4 exec artifacts, which are sequential objects a multi-threaded caller must serialize itself. Liveness is verified nowhere (Section 8 of the main paper).

APPENDIX Q: FRAMEWORK PROBES AND FURTHER OCCURRENCE EVIDENCE

Reading note. The deer-flow bug described below needs no parallelism: it is a sequential blind overwrite under a last-value channel, not a concurrency anomaly, and the main paper does not count it as evidence for any cataloged predicate. It is kept here as a record of what last-value channels do to accumulated state.

Three measurements the argument needs and does not have. (a) *The exposure–latency trade-off of output commit.* Output commit removes exposure only while an effect is held, so what a deployment needs is exposure as a function of the hold deadline, given the latency of its reviews. The synthetic harness fixes one point of that curve (a quarter of reviews arrive after release, and the residue is exactly that quarter, Section 6.3 of the main paper); the live driver that ships with the artifact holds every dependent until the verdict, the opposite extreme, and its runs are not in this paper. The curve between the two points is unmeasured. (b) *The wall-clock study on the real stores*: Appendix O injects its aborts, so the wall-clock price of either discipline under its own abort rate is unmeasured. (c) *A denominator-backed prevalence study*: a sampled population of public LangGraph graphs, classified by whether a reader and a writer of one channel or store key can share a superstep or overlap across threads, with the sampling frame and the classifier's error stated. Every occurrence figure in this paper is either a constructed race or a precondition rate; (c) is the study that would turn them into a prevalence estimate.

CrewAI: the shape is rejected at construction. We probed CrewAI 1.15 the same way and with no model calls (`python/crewai_shape_probe.py`), because the question is what the framework will accept rather than what an agent writes. A crew in which one asynchronous task takes another asynchronous task's output as its context is refused by a model validator before any agent runs. Two controls locate the rule precisely: two asynchronous tasks that share no context edge construct without complaint, so this is not a ban on concurrency; and interposing one synchronous task makes the same pair legal, which is the framework's documented escape and is staging by another name. What the validator rejects is exactly a concurrent read of a concurrently-produced value—the precondition Definition 1 of the main paper requires. CrewAI's memory does not supply one either: its surface is records, matches and slices retrieved by embedding similarity, with no keyed cell for a second agent to overwrite inside a window.

A prediction we tested and must withdraw. An earlier version of this section named the human-in-the-loop `interrupt` as the sharpest remaining case, reasoning that a gate lasting minutes rather than seconds must swamp every latency term and make stale-generation structural rather than a race. We tested it (`python/langgraph_interrupt_probe.py`; deterministic, no model calls, since the question is about resumption semantics rather than about what an agent writes) and the reasoning is wrong. `interrupt` resumes by re-executing the node from its start—the

API documents this—so a read taken before the gate is re-taken after it and the window is closed. That holds where the value lives in a graph channel and where it lives in a `BaseStore`, and identically on `langgraph` 1.2.0, the version of Section 6.1 of the main paper, and 1.2.11.

The mechanism that closes the window opens a different hazard, and the same probe measures it: an irreversible external effect issued *before* the gate is emitted twice, and when the shared value was revised during the gate the two emissions carry different arguments—in our run, a hold for 14 June followed by a hold for 21 June. Three controls separate cause from coincidence: moving the effect after the gate yields one emission, omitting the revision yields two identical emissions, and a graph with no `interrupt` runs each node once. This is duplication, not the within-record reordering A_6 formalizes, and the catalog of Section 3 of the main paper has no predicate for it: the operational model of Section 2 of the main paper assumes at-most-once externalization without ever saying so, and a documented resumption path in the most-deployed framework breaks that assumption. We record it as a limitation of the model rather than enrol it as a fifth anomaly, and note that the fix the framework documents—place external effects after the gate—is itself an instance of the discipline-versus-anomaly trade the lattice is built to express.

Occurrence and susceptibility in deployed frameworks

The anomaly classes of Section 3 of the main paper are not confined to our model: the lowest lattice point L_0—unsynchronized shared mutable state—is the *default* coordination mode of the most widely deployed multi-agent framework, LangGraph, and its anomalies are observable, reproducible, and reported in the wild. We probed this with three complementary, fully reproducible artifacts (deterministic, with no LLM calls, runnable offline; identical results on LangGraph `v1.2.0` and `v1.2.4`). We do not run our Verus-verified detector pipeline against production deployments—that refinement remains the principal follow-up (Section 2 of the main paper)—so what follows is occurrence-and-susceptibility evidence for one framework, not a production-trace prevalence measurement.

Mechanism: the default channel admits the L_0 anomaly class.: LangGraph executes parallel nodes in Pregel supersteps over shared state channels. When a state key carries no reducer (the default), two behaviors arise. (i) Two nodes writing the same key in one superstep fail-stop with `INVALID_CONCURRENT_GRAPH_UPDATE` [56] (“Can receive only one value per step”). (ii) A node that reads a key concurrently updated by another, or a downstream node whose partial update overwrites an accumulated value, silently commits a stale or clobbered result—no error is raised. Our artifact reproduces both on an unmodified LangGraph runtime with deterministic nodes: a parallel edit/summarize graph commits a summary generated from a superseded snapshot (a stale-generation witness in the A_1 family, raising nothing), and a two-writer graph fail-stops. Declaring the key with a merge reducer (e.g. `Annotated[list, op]`) removes the fail-stop and the clobber: a different, stronger channel contract, not a patch. It does *not* remove the stale read—the superstep experiment of Section 6.1 of the main paper runs on a reducer-declared channel and the summary is grounded in the superseded plan in $300/300$ sessions—so the reducer buys lost-update freedom, not L_1. The structural fail-stop check is moreover neither sound nor complete for our A_1 predicate: LangGraph issue #6446 [57] reports it firing when a parallel sub-graph updates a *different* key (a false positive), while the silent variant (ii) shows it is also incomplete. A verified predicate-level specification of the kind we provide (Section 5.1 of the main paper) is a candidate oracle against which such framework checks could be validated.

Occurrence: the detectable variant in real third-party code.: The fail-stop variant raises a fixed error string, so its incidence is measurable. A reproducible GitHub query over issues and pull requests matching the two exact strings LangGraph emits[1] returns **34 distinct third-party repositories** containing an issue or pull request that mentions one of the error strings. They include widely-starred projects (`bytedance/deer-flow`, `FlowiseAI/Flowise`, `CopilotKit/CopilotKit`) alongside a long tail of smaller ones; we report the repository count as an *occurrence* signal only—a star count measures a project’s popularity, not how often the anomaly fires, and a repository mentioning the error string is evidence the fail-stop was reached there, not a prevalence rate. Because the silent variant emits nothing, it is absent from any such count, so **34 is a lower bound on occurrence** and the undetectable variant is the more dangerous of the two.

In the wild: the silent variant, reproduced from a live bug.: The silent variant is not merely a possibility we constructed. ByteDance’s deer-flow carries open issue #3123 [30] (filed May 2026, labeled *help wanted*): a todo list is visible during streaming but *disappears* once the run completes. The project’s own regression test attributes the cause to a downstream node’s partial state update with `todos=None` overwriting the accumulated value under the default last-write-wins channel—a silent lost-update / stale-overwrite in the L_0 class (the surviving state is stale with respect to the accumulated value; we do not claim it instantiates the exact A_1 temporal predicate, only its staleness family). Our artifact reproduces #3123 on an unmodified runtime using deer-flow’s *own* reducer: without it, two seeded todos are silently lost when a downstream node emits `todos=None`; with it, both survive. The fix history is itself instructive about why a consistency lattice is useful: the reducer patch (#3180 [32]) in turn collided with LangChain’s `TodoListMiddleware`, which declares the same `todos` channel with an incompatible type (`Channel 'todos' already exists with a different type`, open issue #3199 [31]); reverting the reducer reintroduces #3123. The resolution is to enforce a single consistent channel contract. The full in-the-wild arc—default $\Rightarrow$ silent loss $\Rightarrow$ reducer ascent $\Rightarrow$ channel-contract conflict $\Rightarrow$ unify the contract—is a production instance of exactly the reasoning the lattice is built to support, including that agreeing the consistency discipline *across components* is the hard part.

Formalized: the reducer fix as a verified lost-update-freedom theorem.: The placements of Section 6.8 of the main paper are informal, but the mechanism deer-flow #3123 exercises we discharge formally. A core Verus model of LangGraph’s per-superstep channel update, parameterized by the reducer, proves (machine-checked) that the additive channel’s value equals the initial value followed by the concatenation of every payload, so no committed contribution is ever lost, while a concrete two-superstep execution exhibits the default last-write-wins channel

1. `INVALID_CONCURRENT_GRAPH_UPDATE` (50 issues/PRs) and “Can receive only one value per step” (56); the union is 44 repositories, of which 34 are third-party (non-`langchain-ai`). Both totals fall below GitHub’s per-query page limit, so the union is complete for these strings rather than a sample. The query and captured results are included in the artifact.

silently dropping a contribution (deer-flow #3123 in miniature) (`lib_langgraph_refinement.rs`; 7 verified, 0 axioms). We are precise about where this sits. Lost update is not one of the four cataloged predicates, and lost-update freedom is not $\neg A_1$: the additive channel preserves every contribution and still hands a co-superstep reader a superseded snapshot (Section 6.1 of the main paper). The theorem is a verified statement about the most-used framework's own channel semantics at the base of the lattice, not a refinement to L_1 as Section 4 of the main paper defines it; an earlier version labeled it $L_0 \to L_1$, and that label is withdrawn. The second construction is the one that reaches L_1: a versioned-channel model under optimistic concurrency (Orleans-style ETag, as AutoGen's persistence layer adopts) proves version monotonicity and that any channel-changing write read the current head—no stale-grounded write, which is read validation and hence the L_1 discipline—with a witness the version-unaware channel accepts but the OCC channel rejects (`lib_occ_l2_refinement.rs`; 8 verified, 0 axioms). That file's header numbers its two disciplines L_1 and L_2 in a local sense, and an earlier version of this paper carried the numbering over as an $L_1 \to L_2$ edge of the lattice; nothing in the file concerns A_3, and that label is withdrawn too. Two channel mechanisms that real frameworks ship are thus verified against their own semantics; the higher edges remain future work.

A second system: Letta shared memory.: The L_0 class is not specific to LangGraph's channel reducers. Letta, the production successor to MemGPT [67], lets several agents share a single memory block, and its own documentation warns that concurrent edits lose updates, advising each agent write a separate section—single-writer partitioning, the oldest remedy for lost updates. We ran the unguarded default: on a self-hosted Letta server, four agents attached to one shared block were asked concurrently to read the shared plan and append their next step (`gpt-4o` backbone, ordinary collaborative-editing prompt; harness `python/letta_a1_probe.py`). Across six rounds of four writers, 18 of 24 appends were silently lost to last-writer-wins. This is the same silent L_0 lost-update as deer-flow #3123, reproduced in an independent deployed system of a different architectural family. We classify it as L_0, *not* a cross-agent A_1 witness, and the reason is architectural rather than observational: Letta re-renders the *current* block into each agent's context at every generation step, so no commit is grounded in a superseded read and Definition 1 of the main paper's precondition cannot persist; the only anomaly concurrent load produces is the write-side clobber we measure. We attempted to certify A_1 by intercepting the model call and report the negative outcome (the per-step refresh leaves no persistent stale read, and modern backbones bypass a drop-in proxy). We read this as architecture-level evidence that the window is a property of the coordination design: refresh-based memory closes it, while the superstep-pinned channels described above do not. A second, independent mechanism closes it the same way—LangGraph's own `interrupt` re-executes a node from its start on resume (Section 8 of the main paper), so a read taken before a human gate is refreshed after it. Two different designs, per-step re-rendering and on-resume re-execution, close the window; superstep pinning leaves it open. The discriminating property is whether a read is retaken, not how long the interval is.

Four mechanisms, one open. Taken with Section 8 of the main paper's CrewAI probe and the AutoGen contrast below, the deployed systems we have examined close the window in four distinct ways and leave it open in one. Letta re-renders the shared block into each agent's context at every generation step. LangGraph's `interrupt` re-executes the node from its start on resume. CrewAI rejects the concurrent-read graph shape at construction. AutoGen's actors own their state and exchange messages, so there is no shared cell. LangGraph's superstep channels pin a read snapshot across a node body that may contain a model call, and are the one case in which the window is open—which is where Section 6.1 of the main paper measures it. We state this as a characterization rather than a survey: five systems is not a sample, and the value of the list is that each entry names a *mechanism* a designer can choose, not a rate. The open empirical question is therefore not whether a deployed framework pins a read snapshot across a long generation—LangGraph's superstep does—but how often deployed graphs place a reader and a writer of one channel in the same superstep, which none of our evidence measures.

Architectural contrast: AutoGen.: The susceptibility tracks the coordination architecture, not the workload. AutoGen uses an actor model: each agent owns its state and agents communicate by asynchronous message passing, so the shared-mutable-state lost-update class does not arise from a built-in channel by construction. The corresponding AutoGen pain is fragmented conversational state—a community discussion [11] describes users building propose/validate/commit layers over the message log (an application-level OCC), and AutoGen's own state-persistence work [13] adopts Orleans-style ETag-based optimistic concurrency. Shared-mutable-state (LangGraph) versus message-passing actors (AutoGen) is precisely the L_0-versus-higher axis the lattice formalizes: the framework that defaults to unsynchronized shared state is the one whose users file the conflicts. We also record one adjacent atomicity gap our model does *not* capture: AutoGen issue #7043 [12] describes a cross-record persistence failure (a completed agent's output is logged but the next-agent queue is not enqueued atomically), related to but outside A_6's within-record reordering; our per-record-atomicity assumption (Section 2 of the main paper) would need relaxing to formalize it.

What this establishes, and what it does not.: The reproductions show that LangGraph's default semantics admit the L_0 class (fail-stop and silent) on an unmodified runtime, that the reducer discipline removes the lost update and not the stale read, that ≥ 34 third-party projects have hit the detectable variant, that deer-flow #3123 is a live silent instance reproduced from the project's own code, that Letta exhibits the same loss in a different architectural family, and—under live inference—that LangGraph's shared store commits stale-grounded bookings in every session in which a concurrent revision lands inside the pinned generation window (Section 6.1 of the main paper). It does *not* establish a production anomaly *rate* (we measure occurrence and susceptibility, not the fraction of runs affected), nor that our model refines either framework in full—only two channel mechanisms are discharged formally—lost-update freedom of the additive reducer and read validation of an Orleans-style ETag channel—the rest being future work. The bounded claim the evidence supports: the lattice's lowest point is the default of the most-used framework, its lost updates are real and reproducible there, and its pinned superstep snapshot admits the stale-generation window (Section 6.1 of the main paper).

APPENDIX R: EXTENDED RELATED WORK

Viotti and Vukolić [75] provide the canonical taxonomy of non-transactional consistency models in distributed storage; their

classification covers per-object, per-key, and session-level guarantees under network propagation delay. Our lattice targets a different operational regime (long-running operations under neural-inference latency, not network propagation), but their methodology—enumerating anomalies and stratifying levels by negation—is the direct analogue we follow. Abadi [7] refines the CAP theorem with a latency/consistency tradeoff that captures, at a higher level of abstraction, the same tension our lattice makes explicit at the level granularity. Globally distributed databases [29], [74] achieve serializability and external consistency under wide-area replication through synchronized clocks and deterministic concurrency control respectively; the techniques bear on the implementability of L_4 guarantees in deployed agent runtimes, modulo cost-asymmetry considerations.

The distributed-systems consistency literature more broadly [22], [61], [63], [69], [73] addresses replicated state but assumes operations are discrete events with summarizable signatures.

Transactional memory and multi-version concurrency control.: Software transactional memory [47], [70] is the closest concurrency-control paradigm to the agent setting: long-running transactions, optimistic execution with abort-and-retry, and conflict detection at commit, in the tradition of optimistic concurrency control [55]. The asymmetry that distinguishes the agent setting from STM is the cost-asymmetry assumption (Section 2 of the main paper): an aborted transaction costs microseconds in STM and seconds-to-minutes in the agent setting, which one would expect to invert the favored strategy from optimistic to pessimistic. Our measurements do not bear that expectation out (Section 6.6 of the main paper): a re-spent generation costs about one generation, validation carries no fixed overhead, and pessimistic locking pays instead in dropped contributions and serialized critical sections. Multi-version concurrency control [20] solves a related problem in databases by retaining multiple versions of each cell to support read-only transactions without conflicts; an MVCC-style mechanism applied to the agent setting would prevent A_1 at the cost of storing pre-generation snapshots for the duration of every operation, which is feasible but expensive at LLM-inference latency. The transaction-modeling literature—specifically Chrysanthis and Ramamritham's [27] ACTA framework for advanced transaction models—provides the formal vocabulary (commit dependencies, abort dependencies, view-serializability) that we adapt informally for the saga-compensation discussion of A_3. Our contribution is not a new transaction model in this lineage but a lattice of consistency predicates over a specific operational regime.

Workflow systems, long-lived transactions, and compensations

The challenge of consistency over long-running activities is not new to multi-agent LLM systems. The advanced-transaction-model literature [28], [36] formalizes long-running and nested-transaction semantics including sagas [39], the original compensation-based approach for long-running activities; the ConTract model [68] extending sagas with explicit dependency contracts; and WS-BusinessActivity for web-services compositions. Modern workflow engines apply these ideas to business-process orchestration. Our work adopts the compensation-aware framing of A_3 from this tradition and the no-write-without-compensation pattern from sagas; the agent setting introduces the additional cost-asymmetry constraint (Section 2 of the main paper) which neither classical sagas nor ConTract address.

Actor systems and message-passing concurrency

The actor model [48] treats concurrent computation as isolated actors communicating via asynchronous messages. Modern implementations—Erlang/OTP, Akka, Microsoft Orleans [19] and its virtual-actor abstraction—realize this model with various strong-consistency extensions including transactional actors, ETag-based optimistic concurrency, and grain-level state. The relevance to multi-agent LLM systems is direct: AutoGen's underlying coordination model is message-passing, and AutoGen's ongoing state-persistence work [13] adopts Orleans-style ETag-based OCC. Our operational model assumes a single shared key-value store, a simplification relative to the message-passing reality of these frameworks; the lattice points and predicates we formalize correspond to the consistency contracts an actor system can implement, but the operational mapping from a message-passing runtime to our shared-store model is non-trivial and deferred to future work.

A related formal lineage is *behavioral* and *session* types [50], which type communication channels by the protocol of messages they carry and statically enforce that interacting components agree on that protocol. This is directly germane to one of our field reports: the deer-flow channel-type conflict ("Channel 'todos' already exists with a different type," issue #3199 [31]), whose resolution is to unify the channel contract across components, is in essence a behavioral-typing concern—components must agree on the consistency discipline of a shared channel before they compose, and multiparty session types are the formal home for that agreement. We note the connection but do not formalize it: our operation-record model classifies runtime traces against a consistency contract rather than statically typing channel protocols, and bridging the two is left to future work.

Verification-aware and model-checking approaches

Two contemporary lines of work address the formal verification of multi-agent LLM systems with methodologies distinct from ours. VeriMAP [77] integrates verification into the planning process itself: a planner agent emits, for each subtask in a DAG decomposition, planner-generated verification functions in Python and natural language; an executor produces JSON outputs verified by these functions, with retries and replanning managed by a coordinator. This is a different design point than the one we occupy: we classify traces post-hoc against a formal consistency contract, whereas VeriMAP integrates per-subtask verification into the plan. The two are complementary; VeriMAP can prevent task-level misalignments that our anomaly detector would not catch (it does not look at subtask-pass criteria), and our detector can identify cross-agent staleness that VeriMAP's per-subtask VFs would not test for (they verify outputs, not inter-agent state coherence).

AgentVerify [9] formalizes agent safety properties in LTL and verifies them via model checking, treating the LLM as a non-deterministic oracle within a finite-state machine defined by the orchestration layer. Its compositional library covers memory integrity, tool-call protocols, MCP/skill invocations, and human-in-the-loop boundaries. The methodological contrast with our work is that AgentVerify uses LTL specifications over orchestration FSMs, whereas we use TLA$^+$ predicates over operation records and lift the executable detectors via Verus refinement. AgentVerify's compositional FSM model is better suited to safety-property verification across complex orchestration topologies; our operation-record model is better suited to anomaly classification on

captured traces. Combining the two methodologies—LTL safety specification over an orchestration FSM with operation-record-level anomaly classification at runtime—is an attractive direction we have not pursued.

Both VeriMAP and AgentVerify descend from a longer tradition of *multi-agent systems verification* that predates LLM agents: temporal-epistemic model checking, embodied in the MCMAS checker [62], verifies CTL/ATL and epistemic properties of interacting agents over an interpreted-systems semantics. That line targets knowledge, strategy, and coordination properties of agent protocols, whereas our predicates target shared-state consistency over operation records; the two are complementary specification layers over the same systems, and we adopt the trace/operation-record granularity precisely because the consistency anomalies we formalize are not naturally expressed as epistemic or strategic modalities.

Runtime verification and trace monitoring

Our detector is, in the terminology of the runtime-verification (RV) field, a *trace monitor*: it decides a formally specified property over an observed execution rather than verifying the producing system. The methodology and its limits are well charted. Havelund and Roşu [44] synthesize monitors from safety properties; Leucker and Schallhart [60] survey the area; and Bauer, Leucker, and Schallhart [18] characterize *monitorability*—which LTL/TLTL properties can be soundly decided from finite prefixes—and give the standard three-valued ($\top/\bot/?$) semantics for online monitoring. Two consequences bear on our detector. First, the anomalies we monitor are finitary predicates over a completed operation history (a non-repeatable read, an aborted predecessor in a causal closure), so they are monitorable in the strong sense—decidable from the trace—which is why the Verus soundness-and-completeness result of Section 5.1 of the main paper is achievable at all, rather than merely a sound one-sided alarm. Second, the RV literature's distinction between detecting a violation and establishing operational impact is exactly the gap our operational-materiality study (Section 6.7 of the main paper) addresses empirically. What our detector adds over generic RV is that the monitor itself is mechanically verified against its specification; most deployed RV tools are unverified monitor implementations. The database-trace checkers Cobra [72] and Elle [10] discussed above are the domain-specific instances of this same monitoring paradigm. Closer to our deployment target, AgentSpec [76] enforces user-written runtime rules over an LLM agent's actions (preconditions on tool calls, budgets, prohibited effects); it is a policy-enforcement layer over single-agent behavior, orthogonal to the shared-state consistency predicates we monitor, and the two compose: AgentSpec rules gate what an agent may do, our detector classifies what a set of agents did to shared state.

Concurrent program logics

The concurrent-semantics lift of Section 5.2 of the main paper sits in the territory of program logics for shared-state concurrency, and we situate it there rather than claim to advance it. The foundational systems—Owicki and Gries's interference-freedom method [66] and Jones's rely-guarantee calculus [51]—reason compositionally about threads sharing memory, and concurrent separation logic (O'Hearn [65], with Brookes's semantics [21]) adds ownership-based local reasoning, mechanized in modern form as Iris [52] and applied to Rust by RustBelt [3]. These logics are the proper tool for discharging—not merely enumerating—the mutex obligation our lift relocates to the runtime's lock crate: a rely-guarantee or CSL proof that `parking_lot::Mutex` realizes the abstract Acquire/Release protocol would close the residual that Section 5.2 of the main paper leaves open, and RustBelt has carried out exactly that style of proof for the standard library's synchronization primitives—which the deployed runtime does not use, so the obligation is not discharged even by proxy. The concurrent store of Section 5.2 of the main paper takes the other route: its lock is `vstd`'s reader–writer lock, verified in the tokenized-state-machine style of Hance et al. [42], [43], so the correspondence is inside the verified library rather than relocated to a program logic. We deliberately do *not* reconstruct such a logic here: our lift is a minimal, mechanically-verified identification of the abstract protocol the sequential refinements require, conditional on that protocol holding, and the program-logic machinery needed to verify the condition is future work (Section 8 of the main paper). Complementing the deductive approach, the stateless-model-checking line for weak memory—GenMC [2] (which we use), CDSChecker [64], and Nidhugg [8]—decides the same lock-protocol question exhaustively but only at bounded thread counts, which is the precise scope and limit of our RC11 check.

Operational transformation and CRDTs for collaborative editing

The collaborative-editing literature treats concurrent edits through operational transformation (OT) and CRDTs. OT, originating in GROVE [34], [35] and deployed in Google Docs, *re-serializes* concurrent operations for the classes its transformation functions cover—tool invocations and budget decrements are not among those classes. CRDTs [69] provide merge functions guaranteeing eventual convergence, with a verification tradition of its own (Gomes et al. [40] verify strong eventual consistency in Isabelle/HOL) parallel to our exclusion-axis verification. CodeCRDT [6] is a CRDT-based agent runtime on a different contract than our L_1, and LangGraph's recommended `operator.add`-reducer workaround for `INVALID_CONCURRENT_GRAPH_UPDATE` [56] is likewise a contract change, not A_1 prevention (Appendix Q). Merge-convergence is thus an axis *orthogonal* to the exclusion lattice (as CALM's coordination-avoidance axis [46] is): for commutative effects a CRDT contract can make A_1 prevention unnecessary, while for non-commutative effects no merge restores a serial order and the exclusion lattice applies. Neither axis subsumes the other; non-serializable convergence points are future work.

Agent-memory systems

The regime our catalog targets—shared mutable state across LLM agents—is realized concretely by agent-memory systems such as MemGPT [67] and its successor Letta, which give agents persistent, editable memory partitioned into context tiers. These systems make shared-mutable-state coordination a first-class concern but provide no explicit consistency contract over concurrent memory edits; they are therefore candidate deployment substrates for the prevention contracts our lattice formalizes, and instrumenting one is the principal empirical follow-up we identify in Section 8 of the main paper. We do not formalize their operational models here.

Probabilistic program verification

The probabilistic refinement of Appendix I stops at a count bound and leaves Hoeffding-grade concentration [49] unmechanized, for

want of real-number probability theory in the current Verus distribution. The probabilistic-verification literature is the natural home for closing it: probabilistic relational Hoare logic [17] for relating a stochastic agent's behavior under counterfactual reads, and weakest-pre-expectation calculi [54] for expectation and concentration reasoning over probabilistic programs; mature probabilistic model checkers such as Storm [33] supply the algorithmic backend such reasoning ultimately invokes. Adapting either to a Verus-style mechanization of the disagreement-probability estimator is the path to a genuine concentration guarantee; we identify it as formalization follow-up rather than claiming it here.

Deterministic databases

Deterministic database systems—Calvin [74], H-Store [53], and the broader line of work on deterministic transaction processing—obtain serializability without locking by ordering transactions globally before execution. The ordering is taken as the canonical schedule and all replicas execute it identically. The connection to our L_4 target is direct: deterministic ordering is one mechanism by which L_4 could be implemented in a multi-agent runtime, with phantom-tool prevention encoded as part of the deterministic schedule. We do not pursue this implementation strategy in the current paper, but identify it as a candidate L_4 runtime worth investigating in deployment.

Two further points of contact are worth drawing out, because they run in opposite directions and the paper is stronger for stating both.

Their premise is one our measurements independently support. CoAgent's case against classical concurrency control is that abort-and-retry destroys the work rather than merely delaying it. Section 6.6 of the main paper reaches the same conclusion from a verified runtime and a different measurement: scored on the joint outcome, pessimistic locking on edit-review is clean in 100 of 100 sessions and complete in 1, and default-SI on triage is complete-and-clean in $45/100$ against the unguarded baseline's $52/100$. Neither result depends on the other's implementation, and they agree that a discipline evaluated only on its anomaly rate will look better than it is.

Where the two part company is what sits inside the correctness argument. MTPO's guarantee is conditional on A3—that a notified agent correctly identifies which of its premises and pending operations a conflicting write invalidates—and that condition is discharged empirically, by a model, at a measured failure rate. The corresponding decision in this development is `can_commit` (Section 6.3 of the main paper), verified two-sided against `commit_valid`, with no model in the trust base and no residual to measure. The two positions are not rivals so much as opposite answers to one question—whether the judgment that a conflict matters should be made by the agent or by the runtime—and the answers have different failure modes: an advisory protocol degrades with model quality, a verified gate degrades with the fidelity of its specification, which Section 6.9 of the main paper enumerates. We take the honest comparison to be that CoAgent buys concurrency at the price of a model-shaped hole in its proof, and the lattice buys a closed proof at the price of refusing work that Section 6.6 of the main paper now quantifies.

Duplicated external effects and checkpoint–restore

The hazard Section 8 of the main paper records—an irreversible effect issued before a re-executable gate and emitted a second time with different arguments—is an instance of the classical *output commit* problem [37], [71]: a system may publish an external effect only when it can guarantee it will not republish it after recovery. Classical rollback-recovery discharges this with identity-keyed duplicate suppression, which is sound because piecewise determinism makes the replayed effect identical to the original. Agent runtimes break that premise from two independent directions. ACRFence [78] documents the first: surveying twelve frameworks it finds tool-call side-effect issues pervasive and re-synthesis nondeterminism defeating request-identifier deduplication. Section 8 of the main paper exhibits the second, which involves no model at all—the shared value is legitimately revised while the gate is open, so the two emissions differ because the *state* changed rather than because the generation did. We record this as a limitation of the operational model rather than as a fifth anomaly, and note that identity-keyed suppression does not repair it: suppressing the second emission is exactly what retains the stale one.

Mechanized consistency conditions

Transactional memory supplies the closest correctness conditions for long-running optimistic operations. Opacity [41] strengthens serializability by constraining what aborted transactions may observe, not only what committed ones leave behind, and a reader who knows that literature will reasonably ask whether L_2 is opacity renamed. It is not: opacity governs the doomed transaction's view, while the L_n predicates govern which committed histories a runtime may emit, and in this regime a doomed operation's view is not the interesting case because re-execution is the dominant cost rather than a correctness hazard. The two are compatible and address different halves of the problem. Chapar [59], discussed in Section 7.5 of the main paper, remains the closest prior mechanization of a consistency model together with a deployed store. Isolde [16] supports tool-assisted reasoning about transactional isolation levels at the specification level, and TxnSails [79] selects isolation levels adaptively from a learned model—the database-side counterpart of the cost-aware refinement Section 8 of the main paper leaves open, and evidence that the selection problem is live in databases as well as in agent runtimes.

Mechanized storage verification, and integrity outside the store

Two lines outside the agent literature bear directly on this work, and a referee is entitled to ask why the paper did not engage them.

The first is the machine-checked concurrent-storage lineage. Perennial [25] extends the Iris concurrency framework with recovery leases, recovery helping and versioned memory, and verifies a concurrent crash-safe mail server in Coq; GoJournal [26] builds a verified concurrent journaling system on Perennial 2.0 and reports finding a serious concurrency bug that the system's unit tests had not, which is the argument for mechanization in one sentence. Closest to our open obligation is Argosy [24], which verifies *layered* storage systems by recovery refinement: exactly the shape of the concrete-store-to-abstract-machine step Section 5.2 of the main paper leaves open, discharged there for a crash model rather than a stale-generation one. The differences are real—those systems verify against crash and durability specifications, ours against an anomaly predicate over an emitted trace, and they work in Coq rather than Verus—but the refinement discipline transfers, and we take Argosy's layering as the template the remaining step should follow rather than inventing one.

The second is a database result this paper is in some sense a restatement of. *Feral concurrency control* [15] studied ORM-backed applications and found that application-level invariants were

used some thirty-seven times more often than transactions, and that a measurable fraction of them fail to hold under concurrency precisely because the store is not told about them. An agent framework is the same arrangement at one further remove: the invariants live in prompts and reducers, the store sees uninterpreted channel writes, and Section 6.4 of the main paper measures one consequence. What is different is that the agent case has no ORM-shaped seam at which a database could be told; the footprint is not merely undeclared but, as Section 6.8 of the main paper records for LangGraph, not statically derivable at all. Read that way, the present work is an attempt to give the feral case a specification it can be checked against.

APPENDIX T: THE DEFINITION-1 TRANSCRIPTION GATE, THE OUTCOME RESCORING, AND THE OBLIGATION-COUNT HISTORY

Three transcriptions of Definition 1. The artifact carries Definition 1 of the main paper in three executable forms: the TLA$^+$ predicate (`Anomalies.tla`), the Verus `spec` restatement (`lib_detector_equivalence.rs::a1_witness`) against which `detect_a1` is proved sound and complete, and the Rust scorer (`rust-analyser/src/anomalies.rs`) that produced the main paper's Tables 3, 4 and 7. They differ in two conjuncts. The Verus `OpRecord` has no agent field, so its predicate omits $h[i].agent \neq h[j].agent$, and it returns "no witness" when either value is absent (`((Some, Some) => r != w, _ => false)`); the Rust scorer also omits the agent conjunct and compares `Option`s, so an absent read value *fires*. Lemma 1 of the main paper (step $\langle 2 \rangle 2$ of `CommitOrder.tla`) shows the agent conjunct is implied by the model's per-agent non-overlap and read-before-commit invariants, so the three forms coincide on every history the model admits. On the committed data they coincide outright: across all 8,425 operation records in the artifact there is no read cell without a recorded value and no same-agent pair whose window is non-empty. `harness/macaudit/predicate_matrix_a1.py --strict` scores every committed cell under all three forms and exits non-zero on any divergence; Table S10 is its output for this version.

The outcome rescoring. The main paper's Table 3 reads the booking's date and the trip date at the end of every live session off the store. The earlier form of that table reported a "material" column computed as $A_1 \wedge (\textit{date_used} \in \textit{text}) \wedge (\textit{final} \notin \textit{text})$, i.e. conditioned on the predicate, which cannot exceed the firing count by construction. The outcome column is `harness/macaudit/outcome_rescore.py`, which reproduces every cell of Table 3 from `results.json` and emits the LaTeX body with `-tex`. Under L_1 the booking disagrees with the final trip date in 34, 7 and 100 of 100 sessions on the three families; in every such session the revision landed after the commit (the committed record carries the revised date in its validating read in the other 66, 93 and 0), so the residue is the post-commit supersession the main paper discusses, and the 100/100 on the local family is the serialization of its serving stack, under which the revision never lands inside the window.

Obligation-count history. Earlier public versions of the artifact stated different Verus totals (274/295, 331/352, 338/359 and 358/379, curated/full) as the development grew between rounds and READMEs were not regenerated; `verus_count.sh` under the pinned build is the only authority, `CLAIMS.json` is written from it, and `harness/macaudit/claims_gate.py` fails the build when any README or `.tex` states a figure that `CLAIMS.json` does not. The per-file inventory of Appendix S is the current one.

APPENDIX U: THE RESIDUE PREDICATE R (ARTIFACT: A_7), THE BASIS LEDGER G_7, AND THE LEDGER-COMPOSITION MATRIX

The predicate in TLA$^+$. The main paper's residue predicate R (Definition 6) is named `SupersededBasisEffect` / `NoA7` in the artifact, where it was introduced as a fifth predicate before the main paper separated it from the anomaly catalog; the artifact names are kept so that scripts and logs match. `Guarded5.tla` extends `Guarded.tla` with two record fields, *basis* (a set of (cell, value) pairs) and *obligation* (a Boolean), one constant G_7, and the predicate

```
SupersededBasisEffect(h) ==
  \E i \in 1..Len(h) :
    /\ h[i].externalized /\ ~h[i].obligation
    /\ \E b \in h[i].basis : \E j \in 1..Len(h) :
        /\ b[1] \in h[j].write_set
        /\ h[j].write_time > h[i].write_time
        /\ \A k \in 1..Len(h) :
             (b[1] \in h[k].write_set
              /\ h[k].write_time > h[i].write_time)
               => h[k].write_time <= h[j].write_time
        /\ h[j].write_values[b[1]] # b[2]
```

In `GCompleteWrite` the basis is a nondeterministic subset of the operation's (cell, read value) pairs—`\E basis \in SUBSET {«c, op.read_values[c]» : c \in op.read_set}`—so every provenance extractor, including the argument-provenance one of Appendix T′ below, is a refinement of the model. G_7 acts at three points and refuses nothing: at commit, an effect released immediately on a basis the store already disagrees with has *obligation* set; a committing write marks, in the same step, every externalized record whose basis holds the written cell with a different value (`Supersedes(k, ws, wv)`); and a release under output commit marks the released record if its basis is stale by then. The invariant checked is `NoA7 == ~SupersededBasisEffect(log)`.

The witness (no guards). TLC reaches a violation of NoA7 at depth five in about three seconds (3,360 distinct states explored): an agent reads $c_1 = \text{NULL}$ (`StartRead`), commits an effect with basis $\{\langle c_1, \text{NULL}\rangle\}$ released at commit (`GCompleteWrite`, *write_time* $= 1$, *externalized*, no obligation), reads again, and commits $c_1 = v_1$ at *write_time* $= 2$. The record at *write_time* $= 2$ is the latest post-commit writer of c_1 and wrote a value other than NULL. The trace is not an A_1 witness (the superseding write is after the first operation's commit, not inside its window). The same depth-five shape with the second operation by the other agent is reached when the agent conjunct is added to the predicate; Definition 6 of the main paper does not add it, because the obligation is owed whoever supersedes. The main paper's verified arm of the live study (probe 303, `-modes verified`) and its bridge (`crates/runtime/examples/l1_server.rs`) are described in Appendix V.

The ledger in Verus. `crates/verus-detector/src/lib_l5_ledger` restates the ledger over the Verus `OpRecord` of Appendix T: a `spec superseded_basis_effect` transcribing `SupersededBasisEffect`, the invariant `inv_l5` (every externalized record whose basis a later committed write disagrees

TABLE S10
The Definition-1 transcription gate on every committed cell: sessions in which the predicate fires under the TLA$^+$ form (Def. 1), the Verus restatement and the Rust scorer. Zero cells diverge.

Cell	n	Def. 1	Verus	scorer
runs/20260913T0424Z edit-review / plan-execute / triage (vanilla)	300	100 / 0 / 42	same	same
runs/20260913T0150Z vanilla (three workloads)	300	100 / 0 / 48	same	same
runs/20260913T0150Z pessimistic (three workloads)	300	0 / 0 / 0	same	same
runs/20260913T0150Z snapshot isolation (three workloads)	300	5 / 0 / 8	same	same
runs/20260913T0128Z vanilla (three workloads)	300	100 / 0 / 52	same	same
live_a1 gpt-4o-mini: vanilla / validated / sequential / superstep	400	60 / 0 / 0 / 42	same	same
live_a1 claude-haiku-4-5: vanilla / validated / sequential / superstep	400	91 / 0 / 0 / 72	same	same
live_a1 Llama-3.2: vanilla / validated / sequential / superstep	400	1 / 0 / 0 / 26	same	same
synthetic pilot traces	700	107	same	same
oprecords / dynamic / MAST / cookbook (width-one windows)	2,170	0	same	same

with carries the obligation flag), four preservation lemmas (commit, superseding write, release under G_3, and the non-superseding write) and `lemma_l5_no_a7` (`inv_l5` implies $\neg R$), plus two non-vacuity lemmas exhibiting a reachable state on each side of the flag. It verifies under the pinned build (`verus-version.txt`): 7 obligations, 0 errors. It is counted as scaffolding and not as a lattice point: the executable ledger of the live study is harness code, so the correspondence between these bytes and the measurement is weaker than at L_1, L_2 and L_4 (main paper Table 2). With it, `verus_count.sh` reports 371 curated and 392 full distinct obligations (Appendix S).

Exhaustive runs with G_7. At the lattice bounds ($|A| = 2$, two cells, one value, *MaxOps* $= 2$, symmetry on agents and cells) NoA7 holds over 5,213,012 distinct states with G_7 alone and over 9,688,601 with G_7 and G_3, where a held effect can be superseded before its release and the release-time registration is the case exercised.

The ledger-composition matrix. `lattice32.sh` runs `Guarded5.tla` for every subset of $\{G_1, G_2, G_3, G_6, G_7\}$ against each of NoA1, NoA2, NoA3, NoA6, NoA7: 160 runs, each expected SAFE (exhaustive) when the matching guard is on and a witness otherwise, with the result, the expectation and the distinct-state count of each run in `lattice32_results.json`. The main paper reads this matrix as a composition result, not as a thirty-two-point lattice: the carrier of G_7 (basis and obligation) is written by no other guard and read by no other predicate, so the sixty-four anomaly cells with G_7 on reproduce `lattice16.sh`'s verdicts, and NoA7's verdict depends on G_7 alone. All 160 runs match their expectation (0 mismatches); the exhaustive runs range over 1,803,848–9,688,601 distinct states, and the sixty-four commit-time cells with G_7 off return the verdicts of `lattice16.sh` one for one. The two-value safety half of `lattice16.sh` on the current model (`lattice16_v2.sh`): all 32 safety runs SAFE, 2.4–17.3M states Both matrices were rerun on the author's machine on 2026-10-14 (TLC 1.8.0): $160/160$ and $32/32$ verdicts as expected, with the same distinct-state counts (`formal/tla/lattice32_results.json`, `lattice16_v2_results.json`).

Appendix T′: A_7 by argument provenance on the live records. `harness/macaudit/a7_rescore.py` scores every committed live session (Table 3 of the main paper) under Definition 6 with the basis recovered from the records alone: for the booking record i (a write to `trip/bookings/*`), $basis(i) = \{(c, read_values_i[c]) : c \in read_set_i,\ read_values_i[c]$ appears in the effect's arguments$\}$, where the arguments are the external write values and, when recorded, the booking text. Nothing is declared by the agent. In the September run no ledger ran, so every *obligation* field is absent and R reduces to "the latest post-commit writer of a basis cell wrote a different value". Over the nine cells (three models $\times$ vanilla, validated, sequential), $A_1 + A_7$ equals the *booking* $\neq$ *final date* count in every cell—`--check` exits non-zero otherwise—and under L_1 the A_1 column is 0 and R equals the residue (34, 7 and 100 of 100). The L_5 arm of probe 303 (`--modes ledger`) runs the same harness with a basis ledger (`BasisLedger.register / superseded`) wrapped around the store and a rebooking compensator that issues a replacement booking when an obligation is registered; the scorer reads its *obligation* and *compensation* fields, so a correct ledger yields $R = 0$ by construction and the outcome column measures the compensator. In the October run (main paper Table 3) the ledger arm's outcome is booking $\neq$ final $0/100$ on all three families (5, 45, 53 obligations, all discharged). `a7_rescore.py --check` now also scores every probe-303 trace (a trace is one seed-to-booking run; the offset sweep appends every offset's run of a session to one file, each opening with its own seed record) and compares A_1, R and *booking* $\neq$ *final* with the cell the probe wrote: $39/39$ cells agree (three families $\times$ four arms at 150 ms, plus three $\times$ two modes $\times$ four offsets). Under the ledger arm the records carry no *obligation* field; the ledger registers the obligation by firing the compensator, so the scorer reads an obligation off the later compensation record on the same effect cell—the probe's own count and the independent one agree on $R = 0$ in all three ledger cells. After a discharge the harness ledger re-arms the entry with the compensated basis, so a further revision would register a further obligation; this workload revises once per session, and that path is not exercised by Table 3. `hypothesis_gates.py` on the same 9,649 records (3,600 traces): H0 and H1 hold (0 violations), value-unique writes hold, 418 A_1 witnesses, none overwrite-restore, basis recovered for $3{,}600/3{,}600$ bookings; the totals in the main paper add these to the September campaigns. (An earlier version of the gate split a sweep file by session only, merging the four offset runs of a session into one trace and reporting 1,014 spurious same-agent overlaps; the trace delimiter is now the seed record, and the probe tags every record with its offset.)

The offset sweep (main paper Fig. 3). Table S11 lists every cell of the sweep: revision probability 0.5, $n = 100$ per cell, the date agent started 0, 150, 500 or 1,500 ms after the flight agent's read.

TABLE S11
Probe 303 offset sweep, per cell: A_1 / R / booking $\neq$ final (of 100 sessions; about half revise). The three columns of each cell sum as $A_1 + R =$ wrong in every cell.

Model	Mode	0 ms	150 ms	500 ms	1,500 ms
gpt-4o-mini	vanilla	40/11/51	36/15/51	1/43/44	0/44/44
	validated (L_1)	0/4/4	0/13/13	0/40/40	0/42/42
claude-haiku-4-5	vanilla	46/5/51	47/4/51	4/40/44	0/44/44
	validated (L_1)	0/2/2	0/4/4	0/46/46	0/42/42
deepseek-chat	vanilla	39/12/51	20/31/51	2/42/44	0/44/44
	validated (L_1)	0/17/17	0/26/26	0/52/52	0/42/42

APPENDIX V: CLOSEST SYSTEMS, DIMENSION BY DIMENSION; THE VERIFIED-STORE BRIDGE

The verified-store bridge. The live LangGraph study's L_1 arm re-implements read validation at the application layer (re-read, discard, regenerate), which measures the predicate–harm gap but does not execute the verified bytes. Probe 303's `verified` mode routes the same decision through `si_concurrent::SiConcurrent`, the Verus-verified L_1 store of the runtime crate, running as a subprocess (`crates/runtime/examples/l1_server.rs`, JSON lines on stdio): every LangGraph `get` is paired with a verified `begin` and every `put` with a verified `commit` under one harness mutex, a refused commit regenerates, and the per-session verified trace is kept in the receipt. The bridge is about 100 lines of Rust plus the harness's interning of cell names and values to integers; it and the mutex are the only additions to the trust base (item xi of the main paper), and the validation decision itself—`validate` / `commit_step`—is the verified code. A protocol mock (`probes/303_l1_server_mock.py`) exercises the client without a Rust toolchain and is not a measurement path.

APPENDIX S: VERIFIED OBLIGATION INVENTORY

The verification sections report 371 curated and 392 full distinct verified proof obligations (0 errors). This appendix reproduces the per-file accounting those figures derive from, exactly as emitted by `verus_count.sh`. The script verifies every `src/*.rs` file in the `verus-detector` crate standalone under `verus -crate-type=lib`, subtracts each obligation that one file re-verifies from another once per re-inclusion edge, and prints the arithmetic, so the totals are auditable rather than asserted. The default invocation applies the curated exclusion of four non-headline helper files (marked † in Table S13); `verus_count.sh -full` clears that exclusion. Every counted file verifies at 0 errors.

Re-inclusion edges. `lib_consistency_lattice.rs` re-includes `lib_l2_safety.rs` (−33, via `mod`), `lib_l3_safety.rs` (−6, `mod`), and `lib_l4_safety.rs` (−9, `mod`); `lib_l2_exec.rs` re-includes the L_2 model textually (−33). These four edges remove the 81 double-counted obligations. `lib_l2_safety.rs` is subtracted twice— once for each distinct parent that re-includes it—leaving its 33 obligations counted exactly once.

Totals. The full distinct total is $473 - 81 = 392$. The four † helper files contribute $9+5+3+4 = 21$ obligations to the raw sum; excluding them gives a curated raw sum of 452 and hence the curated distinct total $452 - 81 = 371$ (both under the pinned build `0.2026.09.13.671956e`; the previous build counted 364/385 before `lib_l5_ledger.rs` existed). `verus_count.sh` reproduces the latter; `verus_count.sh -full` reproduces the former.

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TABLE S12
The systems nearest this paper, on the dimensions a reader needs to compare guarantees. "Model" = a checked formal model; "proof" = machine-checked proof; "bytes" = verified code executed in the evaluation. Entries are our reading of each paper's own statements; "—" = not addressed.

System	Consistency model (history property)	Read validation before release	Tool validity at dispatch	External effects	Formal verification	Implementation correspondence	Evaluation; cost
This paper	four named anomalies; CORC (Thm. 1); residue R	L_1 at commit; validated-at-dispatch L_4	id and signature (L_4)	held until causal basis externalized (L_2); sequenced (L_3); obligation ledger for R	TLC (bounded), TLAPS (Thm. 1), Verus (detectors, L_1–L_4)	bytes at L_1 (pilot, verified arm), L_2, L_4; port at L_3	controlled races, synthetic cascades, pilot; per sound committed op
MemTX [80]	two invariants (belief commit)	staged writes under SI	—	gated on current belief; cascading repair on retraction	bounded enumeration (5.5M states)	not stated	five backbones; per session
CoAgent [81]	advisory protocol	notify-and-repair	—	—	—	—	model-judged repair (5% failure); lock/OCC cost
STORM [82]	per-file read validation	yes (L_1-like)	—	—	—	—	file workloads
Agentic transactions [83]	semantic ACID (single agent)	explore–execute–validate cycles	skill hub	transaction-aware state	—	—	task success (+10.6%)
CommitGuard [84]	commit-time authorization	evidence fresh at commit	—	fail-closed monitor at commit	—	—	protected commit points
Stale $\neq$ unsafe [85]	semantic commit predicates	freshness guards (measured 92–95% benign blocks)	—	—	—	—	guard precision under races
EffectMatch [86]	approved-outcome validation	—	—	persistent outcomes compared with approval	—	—	206 business tasks
Versioned execution [87]	version holds publish authority	—	—	obsolete version revoked	—	vLLM implementation	TTFT
SagaLLM [88]	saga compensation	—	—	compensating actions	—	—	planning tasks
Atomix [89]	atomic multi-step	—	—	—	—	—	—

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TABLE S13
Per-file Verus obligation counts (`verus_count.sh -full`). † marks the four helper files excluded from the 371 curated headline; the default `verus_count.sh` run applies that exclusion. `verified_a1.rs` carries no proof obligations. The distinct totals subtract re-included obligations once per edge (listed below the table).

File (`verus-detector/src/`)	Verified
`lib_a3_residue.rs`	6
`lib_a4_split_view.rs`	9
`lib_concurrent_semantics.rs`	9
`lib_consistency_lattice.rs`	53
`lib_default_si.rs`	9
`lib_detect_a1_exec.rs`†	9
`lib_detector_equivalence.rs`	24
`lib_l2_exec.rs`	73
`lib_l2_projection.rs`	3
`lib_l2_safety.rs`	33
`lib_l3_exec.rs`	7
`lib_l3_safety.rs`	6
`lib_l3_sequencer.rs`	5
`lib_l4_exec.rs`	11
`lib_l4_safety.rs`	9
`lib_l5_ledger.rs`	7
`lib_langgraph_refinement.rs`	7
`lib_occ_l2_refinement.rs`	8
`lib_pessimistic_exec.rs`†	5
`lib_pessimistic_invariant.rs`†	3
`lib_refinement_default_si.rs`	18
`lib_refinement_pessimistic.rs`	31
`lib_refinement_ssi_chain.rs`	19
`lib_refinement_ssi.rs`	18
`lib.rs`	23
`lib_rustbelt_interface.rs`	4
`lib_si_commit_invariant.rs`†	4
`lib_pess_concurrent.rs`	31
`lib_si_concurrent.rs`	18
`lib_si_projection.rs`	3
`lib_ssi.rs`	8
`verified_a1.rs`	—
Raw per-file sum (counted files)	473

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